Practicing Dynare:
A Fiscal Policy Focus

02 November 2026
Work-in-Progress
Not for Distribution

Abstract

This document updates *Practicing Dynare* by Barillas et al. (2010), with a focus on fiscal-policy applications drawn from *Recursive Macroeconomic Theory* by Ljungqvist and Sargent. It develops stochastic growth models and deterministic fiscal experiments through model equations, calibration and corresponding Dynare code. The examples cover approximation, simulation, estimation and the effects of anticipated and unanticipated policy changes. A further section is reserved for a full-scale flexible-price fiscal model. The aim is to help readers understand model solutions, interpret economic responses and construct their own policy experiments.

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1 Introduction

This guide updates the examples in Barillas et al. (2010) and develops their fiscal-policy applications using Ljungqvist and Sargent's *Recursive Macroeconomic Theory*. Sections 2 and 3 introduce stochastic growth models, approximation, simulation and estimation. Sections 4 and 5 examine government spending and taxation in deterministic growth models, including anticipated policy changes, bond yields and elastic labour supply. Section 6 is reserved for a full-scale flexible-price fiscal model.

Each example links the model equations and calibration to a runnable Dynare file. The files in `Code/sections` present individual exercises; those in `Code/full_replications` produce the corresponding section outputs. Appendix A lists these files. Accompanying notes explain less common commands and calculations.

In fiscal simulations, simulation period 1 represents model date $t = 0$. A policy change at $t = 10$ therefore begins in simulation period 11.

2 The Neoclassical growth model

2.1 The model

Capital available at the beginning of period t is k_{t-1}; capital chosen during t for use at $t+1$ is k_t. This coincides with Dynare timing: `k(-1)` is k_{t-1} and `k` is k_t.

Model 2.1: variable definitions, units and baseline calibration.

Symbol	Code	Meaning / calibration
c_t	`c`	Consumption; endogenous.
k_{t-1}	`k(-1)`	Physical capital available at the beginning of period t; predetermined.
k_t	`k`	Capital chosen during period t for use in period $t+1$; endogenous.
l_t	`lab`	Hours as a fraction of available time, between zero and one; endogenous.
z_t	`z`	Stationary log productivity; endogenous AR(1) state.
ϵ_t	`e`	Exogenous independent Gaussian innovation, mean zero and standard deviation $\sigma = 1$.
β	`bet`	Discount factor: 0.987.
θ	`the`	Consumption share in the consumption–leisure utility composite: 0.357.
δ	`del`	Depreciation rate: 0.012.
α	`alp`	Capital share: 0.4.
τ	`tau`	Utility curvature: 2; this is not a fiscal tax rate.
ρ	`rho`	Persistence of log productivity: 0.95.
s	`s`	Multiplier on the innovation in productivity: 0.007.
σ	`stderr e`	Innovation standard deviation: 1. The effective productivity innovation SD is $s\sigma = 0.007$.

Capital is predetermined in production. All calibration values are per model period; the numbers in `initval` are numerical guesses, not additional calibrated parameters.

We study a widely used stochastic neoclassical growth model with leisure (see, for example, Cooley and Prescott (1995)). A representative household's problem is

$$\max_{\{c_t,l_t\}_{t=0}^{\infty}} E_0 \sum_{t=1}^{\infty} \beta^{t-1} \frac{\left(c_t^{\theta}\left(1-l_t\right)^{1-\theta}\right)^{1-\tau}}{1-\tau}$$

subject to the resource constraint

$$c_t + i_t = e^{z_t} k_{t-1}^{\alpha} l_t^{1-\alpha} \tag{1}$$

the law of motion for capital

$$k_t = i_t + (1-\delta)k_{t-1} \tag{2}$$

and the stochastic process for productivity

$$z_t = \rho z_{t-1} + s\epsilon_t \tag{3}$$

with $\epsilon_t \sim N\left(0, \sigma^2\right)$.
The system of equations characterizing an equilibrium is comprised of equations (1), (2) and (3), the Euler intertemporal condition

$$\frac{\left(c_t^{\theta}\left(1-l_t\right)^{1-\theta}\right)^{1-\tau}}{c_t} = \beta E_t \left[\frac{\left(c_{t+1}^{\theta}\left(1-l_{t+1}\right)^{1-\theta}\right)^{1-\tau}}{c_{t+1}} \left(1 + \alpha e^{z_{t+1}} k_t^{\alpha-1} l_{t+1}^{1-\alpha} - \delta\right) \right] \tag{4}$$

and an optimality condition for supply of labor

$$\frac{1-\theta}{\theta}\frac{c_t}{1-l_t} = (1-\alpha)e^{z_t} k_{t-1}^{\alpha} l_t^{-\alpha}. \tag{5}$$

Use (1) and (2) to get

$$k_t = e^{z_t} k_{t-1}^{\alpha} l_t^{1-\alpha} - c_t + (1-\delta)k_{t-1}. \tag{6}$$

An equilibrium is characterized by a system of four equations (3), (4), (5) and (6).

2.2 Calibration and approximation

2.2.1 Calibration

To estimate the policy function, parameters that enter the model are calibrated as in Table 1. For more on the calibrations of the related parameters, please see, for example, Cooley and Prescott (1995).

2.2.2 Approximation

The following file implements the model in Dynare. The stochastic Euler equation uses next-period marginal product, consistent with equations (1)–(6).

```
// ------------------------------------------------------------------------
// Section 2.2 | One-sector growth | Approximation
// ------------------------------------------------------------------------
// 1. Variables and parameters
// c: consumption; k: capital chosen today; lab: hours; z: log productivity.
var
    c      // Consumption
    k      // Physical capital chosen today for use next period
    lab    // Labour supplied (hours)
    z      // Log productivity
;
varexo
    e    // Standardized productivity innovation
;

parameters bet the del alp tau rho s;
bet = 0.987;    // Discount factor
the = 0.357;   // Consumption share in the consumption-leisure composite
del = 0.012;   // Depreciation rate
alp = 0.4;     // Capital share
tau = 2;       // Curvature of utility
rho = 0.95;    // Persistence of productivity
s   = 0.007;   // Scale multiplying the productivity innovation

// 2. Equilibrium conditions
// k(-1) is the capital available today; k is used in production tomorrow.
model;
    // Euler equation: today's marginal utility equals the expected
    // discounted marginal utility of next period's capital payoff.
    (c^the * (1-lab)^(1-the))^(1-tau) / c =
        bet * (c(+1)^the * (1-lab(+1))^(1-the))^(1-tau) / c(+1)
        * (1 + alp*exp(z(+1))*k^(alp-1)*lab(+1)^(1-alp) - del);

    // Intratemporal condition: consumption versus leisure.
    c = the/(1-the) * (1-alp) * exp(z) * k(-1)^alp
        * lab^(-alp) * (1-lab);

    // Resource constraint and capital accumulation.
    k = exp(z)*k(-1)^alp*lab^(1-alp) - c + (1-del)*k(-1);

    // Stationary log-productivity process.
    z = rho*z(-1) + s*e;
end;

// 3. Initial guesses; steady solves for the actual steady state.
initval;
    k   = 10;
    c   = 0.7;
    lab = 0.3;
```

```
    z   = 0;
end;

shocks;
    var e; stderr 1;
end;

steady;
check;

// 4. Approximation and artificial data
// Fixing the seed makes repeated runs use the same random innovations.
set_dynare_seed(2025);
// nograph suppresses figures; the computed IRFs remain in oo_.irfs.
stoch_simul(order=2, periods=1000, drop=100, irf=40, nograph);

// Export the observed series for the estimation example.
// verbatim passes ordinary MATLAB/Octave code through the Dynare parser.
// strcmp selects a row by name; the apostrophe makes a column of data.
// verbatim copies this block unchanged into the generated MATLAB/Octave
// program. It is not a Dynare equilibrium block; use % comments inside.
verbatim;
c = oo_.endo_simul(strcmp(M_.endo_names, 'c'), :)';
save('s02_simudata.mat', 'c');
end;
```

Parameter	Calibration
β	0.987
θ	0.357
δ	0.012
α	0.4
τ	2
ρ	0.95
s	0.007
σ	1

Table 1: Parameters' calibration

Notes: One-sector growth model: β discounts utility, θ is the consumption share, δ depreciation, α the capital share, τ utility curvature, ρ log-productivity persistence, s the innovation multiplier and σ the innovation SD. Rates are per model period.

Table 2 gives the second-order decision rules at the calibration above. For each column, multiply the coefficients by the terms in the first column and add them to the constant. For example, a unit increase in beginning-of-period capital raises consumption by approximately 0.027972 goods at first order, holding productivity and the innovation fixed.

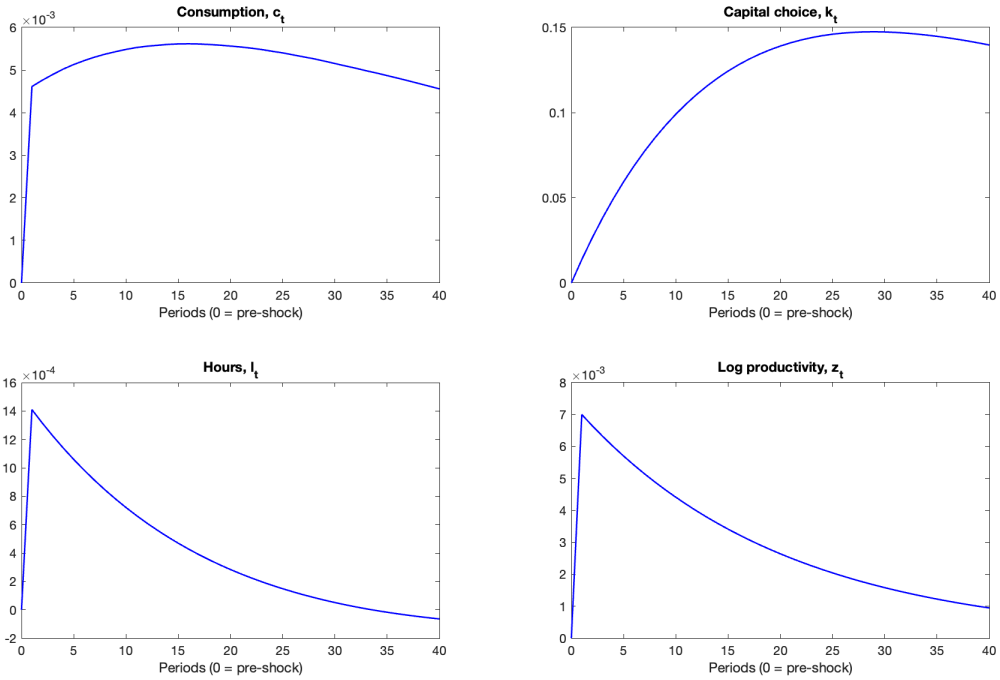


Figure 1: Responses to a productivity innovation in the one-sector model.
Notes: c_t is consumption, k_t the end-of-period capital choice, l_t hours and z_t log productivity. The shock has effective standard deviation 0.007. Responses are changes in model units; period 0 is the pre-shock reference.

Term	c_t	k_t	l_t	z_t
Constant	1.491629	29.288977	0.291594	0.000000
Correction (included)	-0.000002	0.000004	0.000000	0.000000
$\widehat{k}_{t-1}$	0.027972	0.978257	-0.001831	0.000000
z_{t-1}	0.622874	1.853511	0.191285	0.950000
e_t	0.004590	0.013657	0.001409	0.007000
$\widehat{k}_{t-1}^2$	-0.000182	-0.000086	0.000025	0.000000
$z_{t-1}\widehat{k}_{t-1}$	0.007307	0.024441	0.000712	0.000000
z_{t-1}^2	0.237402	1.157824	-0.008033	0.000000
e_t^2	0.000013	0.000063	0.000000	0.000000
$\widehat{k}_{t-1}e_t$	0.000054	0.000180	0.000005	0.000000
$z_{t-1}e_t$	0.003499	0.017063	-0.000118	0.000000

Table 2: Policy and transition coefficients for the one-sector model.
Notes: c_t is consumption, k_t capital chosen during t, l_t hours and z_t log productivity. $\widehat{k}_{t-1} = k_{t-1} - \bar{k}$ is beginning-of-period capital less its deterministic steady state; productivity has steady state zero. The innovation e_t has standard deviation one and enters productivity multiplied by $s = 0.007$. The constant already includes the reported second-order correction; do not add the correction row again. Squared-term coefficients already include the Taylor-expansion factor $1/2$. Values are rounded to six decimals.

2.3 Estimation

The simulated data series for consumption are used to estimate some unknown parameters of the model. Suppose we would like to estimate the preference parameters θ and τ, and the stochastic process for productivity, summarized by two parameters ρ and σ. The following code instructs Dynare to conduct the estimation.

```
// ------------------------------------------------------------------------
// Section 2.3 | Bayesian estimation
// ------------------------------------------------------------------------
// 1. Variables and parameters
// c: consumption; k: capital chosen today; lab: hours; z: log productivity.
var
    c      // Consumption
    k      // Physical capital chosen today for use next period
    lab    // Labour supplied (hours)
    z      // Log productivity
;
varexo
    e    // Standardized productivity innovation
;

parameters bet the del alp tau rho s;
bet = 0.987;    // Discount factor
the = 0.357;   // Consumption share in the consumption-leisure composite
del = 0.012;   // Depreciation rate
alp = 0.4;     // Capital share
tau = 2;       // Curvature of utility
rho = 0.95;    // Persistence of productivity
s   = 0.007;   // Scale multiplying the productivity innovation

// 2. Equilibrium conditions
// k(-1) is the capital available today; k is used in production tomorrow.
model;
    // Euler equation: today's marginal utility equals the expected
    // discounted marginal utility of next period's capital payoff.
    (c^the * (1-lab)^(1-the))^(1-tau) / c =
        bet * (c(+1)^the * (1-lab(+1))^(1-the))^(1-tau) / c(+1)
        * (1 + alp*exp(z(+1))*k^(alp-1)*lab(+1)^(1-alp) - del);

    // Intratemporal condition: consumption versus leisure.
    c = the/(1-the) * (1-alp) * exp(z) * k(-1)^alp
        * lab^(-alp) * (1-lab);

    // Resource constraint and capital accumulation.
    k = exp(z)*k(-1)^alp*lab^(1-alp) - c + (1-del)*k(-1);

    // Stationary log-productivity process.
    z = rho*z(-1) + s*e;
end;

// 3. Initial guesses; steady solves for the actual steady state.
initval;
    k   = 10;
```

```
    c   = 0.7;
    lab = 0.3;
    z   = 0;
end;

shocks;
    var e; stderr 1;
end;

steady;
check;

// 4. Bayesian estimation
// First run s02_02_growth_approx.mod in this folder.
// Each row gives a starting value, prior family, prior mean and prior SD.
// Section 2 fixes s=0.007 and estimates stderr(e), not both shock scales.
varobs c;

estimated_params;
    stderr e, 1, inv_gamma_pdf, 1, 0.2;
    rho, 0.95, beta_pdf, 0.93, 0.02;
    the, 0.357, beta_pdf, 0.3, 0.05;
    tau, 2, normal_pdf, 2.1, 0.3;
end;

set_dynare_seed(2025);
// mode_compute=4 searches for the posterior mode before sampling.
// A short teaching chain; increase mh_replic for empirical inference.
estimation(datafile=s02_simudata, mode_compute=4, mh_replic=1000,
           mh_nblocks=1, nograph, nodisplay);
```

The prior distributions used in the estimation file are listed below. Beta priors keep the persistence and consumption-share parameters between zero and one. The inverse-gamma prior restricts the innovation standard deviation to positive values. Inspect the posterior summaries produced by your run; the short teaching chain is an introduction to the workflow, not evidence of MCMC convergence.

Parameter	Distribution	Mean	Std.Dev.
ρ	Beta	0.93	0.02
θ	Beta	0.3	0.05
τ	Normal	2.1	0.3
σ	Inv. Gamma	1	0.2

Table 3: Priors

Notes: One-sector priors: ρ is productivity persistence, θ the consumption share, τ utility curvature and σ the standard deviation of innovation e. Std.Dev. denotes the prior standard deviation. The productivity multiplier $s = 0.007$ is fixed.

Table 4 summarizes the posterior from the simulated-data exercise. The posterior mean describes the centre of the retained draws; the 90% highest-posterior-density (HPD) interval

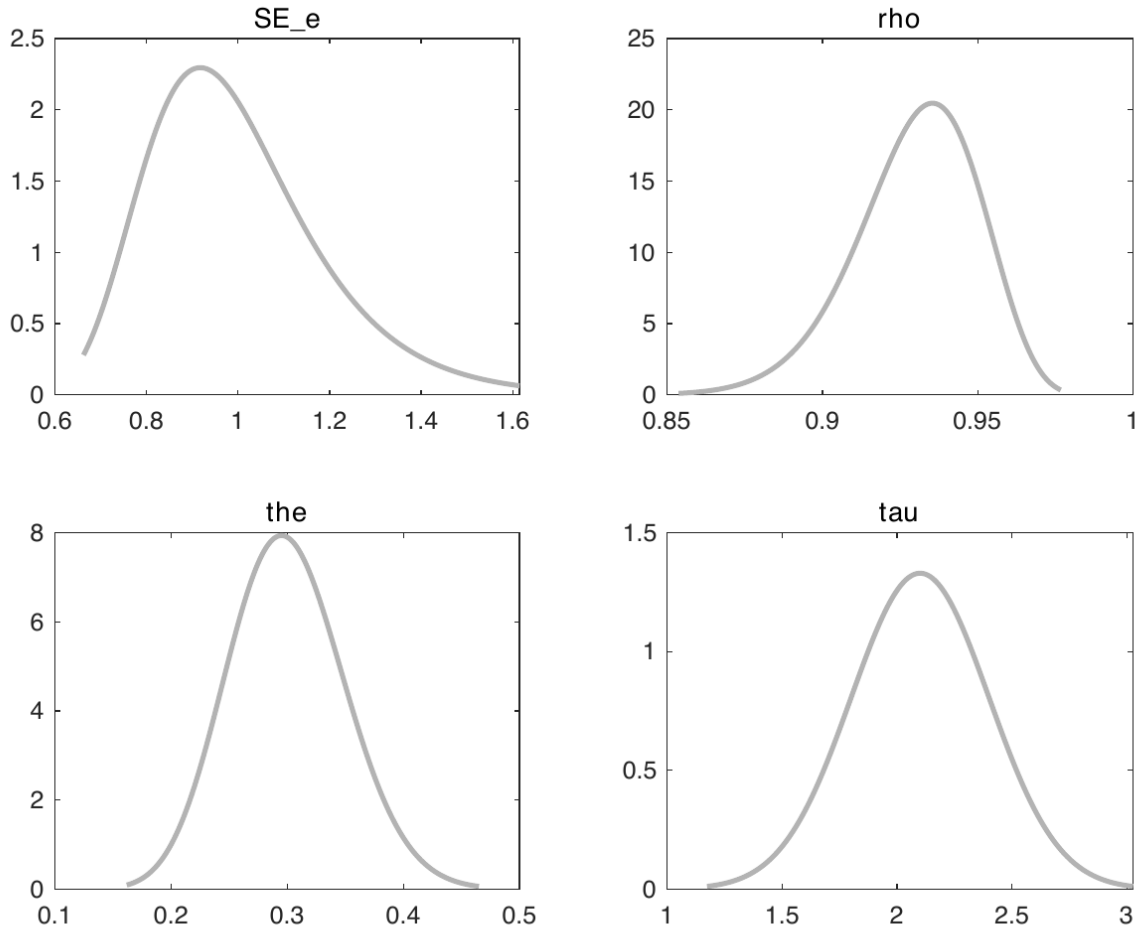


Figure 2: Prior distributions for the one-sector model.
Notes: ρ is productivity persistence, θ the consumption share, τ utility curvature and σ the innovation standard deviation. Horizontal axes are parameter values and vertical axes are probability densities.

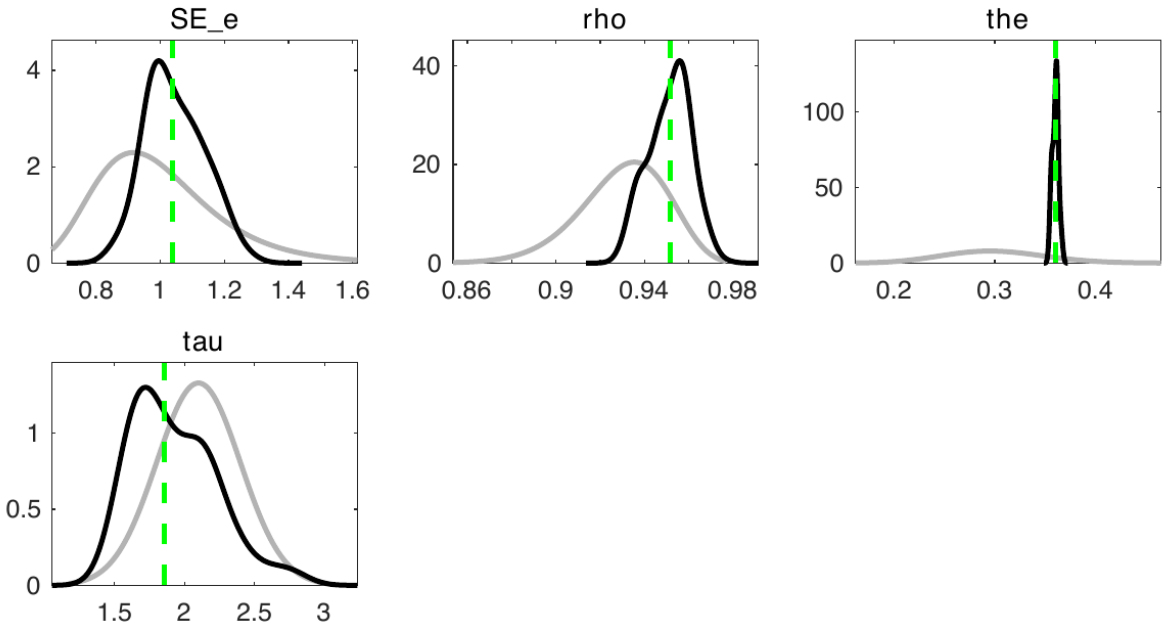


Figure 3: Prior and posterior distributions for the one-sector model.

Notes: ρ is productivity persistence, θ the consumption share, τ utility curvature and σ the innovation standard deviation. Horizontal axes are parameter values and vertical axes are probability densities. The posterior uses simulated observations and a 1,000-draw teaching chain; convergence must be assessed before drawing empirical conclusions.

summarizes their uncertainty.

Parameter	Prior mean	Posterior mean	90% HPD interval	
ρ	0.930	0.9512	0.9358	0.9645
θ	0.300	0.3602	0.3552	0.3646
τ	2.100	1.9265	1.4895	2.3129
σ	1.000	1.0450	0.9361	1.2061

Table 4: Posterior summary for the one-sector model.
Notes: ρ is productivity persistence, θ the consumption share, τ utility curvature and σ the standard deviation of innovation e_t; the productivity multiplier $s = 0.007$ is fixed. Priors are listed in Table 3. Results use simulated consumption and a 1,000-draw teaching chain. This short chain illustrates Bayesian estimation; its intervals do not establish MCMC convergence.

3 International Business Cycle Model

We study a simplified version of the two-country production model in Kim and Kim (2003).

3.1 The model

Model 3.1: country variables, log coding and baseline calibration.

Symbol	Code	Meaning / calibration
$C_{i,t}$	`c1`, `c2`	Country $i = 1, 2$ consumption in levels; code variables are $c_i = \ln C_i$.
$K_{i,t-1}$	`k1(-1)`, `k2(-1)`	Beginning-of-period physical capital in each country; code variables are its logs.
$K_{i,t}$	`k1`, `k2`	Capital chosen during t for use in $t + 1$; code variables are its logs.
$A_{i,t}$	`a1`, `a2`	Country productivity in levels; code uses $a_i = \ln A_i$.
$\epsilon_{i,t}$	`e1`, `e2`	Independent mean-zero Gaussian country innovations; both SDs are 0.08.
β	`beta`	Discount factor: 0.98.
γ	`gamma`	Consumption curvature: 2.
δ	`delta`	Depreciation rate: 0.05.
α	`alpha`	Capital share: 0.4.
ρ	`rho`	Country productivity persistence: 0.85.

Countries are identical ex ante and markets are complete. Consumption and capital are endogenous; the innovations are exogenous. The code uses log variables, while the equations below use levels. Initial values supplied in code are log guesses.

Two countries are identical ex-ante and markets are complete. A solution to a Pareto planner's problem is characterized by the following equations

$$
\begin{aligned}
C_{1,t} &= C_{2,t} \\
C_{1,t}^{-\gamma} &= \beta \mathbb{E}_t C_{1,t+1}^{-\gamma} \left(\alpha A_{1,t+1} K_{1,t}^{\alpha-1} + 1 - \delta \right) \\
C_{2,t}^{-\gamma} &= \beta \mathbb{E}_t C_{2,t+1}^{-\gamma} \left(\alpha A_{2,t+1} K_{2,t}^{\alpha-1} + 1 - \delta \right) \\
A_{1,t} K_{1,t-1}^{\alpha} + A_{2,t} K_{2,t-1}^{\alpha} &= C_{1,t} + C_{2,t} + K_{1,t} - (1-\delta) K_{1,t-1} + K_{2,t} - (1-\delta) K_{2,t-1} \\
\ln A_{1,t+1} &= \rho \ln A_{1,t} + \epsilon_{1,t+1} \\
\ln A_{2,t+1} &= \rho \ln A_{2,t} + \epsilon_{2,t+1}
\end{aligned}
$$

The first three equalities are the first-order conditions with regard to the consumption ($C_{i,t}$), and the fourth one is the world resource constraint. The last two describe the law of motion for the technological shock $(A_{i,t})$.$K_{i,t-1}$ denotes the capital stock of country i at time t.

3.2 Calibration and approximation

3.2.1 Calibration

The following parameter values are used.

Parameter	Calibration
β	0.98
δ	0.05
α	0.4
ρ	0.85
σ_{ϵ_1}	0.08
σ_{ϵ_2}	0.08

Table 5: Parameter Values
Notes: Two-country growth calibration: β discounts utility, δ is depreciation, α the capital share, ρ productivity persistence and σ_{ϵ_i} country innovation SDs. Consumption curvature is $\gamma = 2$. Rates and innovation variances are per model period.

Dynare computes the policy function as a second-order approximation around the (log) steady-state. The .mod Dynare file of the system is as follows.

```
// ------------------------------------------------------------------------
// Section 3.2 | Two-country growth | Approximation
// ------------------------------------------------------------------------
// 1. Variables and parameters
// All endogenous variables are logs. Countries share consumption under
// complete markets; each country has its own capital and productivity.
var
    c1    // Log consumption in country 1
    c2    // Log consumption in country 2
    k1    // Log capital chosen in country 1
    k2    // Log capital chosen in country 2
```

```
    a1    // Log productivity in country 1
    a2    // Log productivity in country 2
;
varexo
    e1    // Productivity innovation in country 1
    e2    // Productivity innovation in country 2
;

parameters gamma delta alpha beta rho;
gamma = 2;       // Consumption curvature
delta = 0.05;   // Depreciation rate
alpha = 0.4;    // Capital share
beta  = 0.98;   // Discount factor
rho   = 0.85;   // Productivity persistence

// 2. Equilibrium conditions
// exp(k1(-1)) and exp(k2(-1)) are the capital stocks used today.
model;
    // Complete risk sharing between identical households.
    c1 = c2;

    // Euler equations for investment in each country.
    exp(c1)^(-gamma) = beta * exp(c1(+1))^(-gamma)
        * (alpha*exp(a1(+1))*exp(k1)^(alpha-1) + 1-delta);

    exp(c2)^(-gamma) = beta * exp(c2(+1))^(-gamma)
        * (alpha*exp(a2(+1))*exp(k2)^(alpha-1) + 1-delta);

    // World output finances consumption and net capital accumulation.
    exp(c1) + exp(c2) + exp(k1) - (1-delta)*exp(k1(-1))
        + exp(k2) - (1-delta)*exp(k2(-1)) =
        exp(a1)*exp(k1(-1))^alpha + exp(a2)*exp(k2(-1))^alpha;

    // Country-specific productivity processes.
    a1 = rho*a1(-1) + e1;
    a2 = rho*a2(-1) + e2;
end;

// 3. Initial guesses in logs, and innovation standard deviations.
initval;
    k1 = 2.8;
    k2 = 2.8;
    c1 = 0.8;
    c2 = 0.8;
    a1 = 0;
    a2 = 0;
end;

shocks;
    var e1; stderr 0.08;
    var e2; stderr 0.08;
end;

steady;
```

```
check;

// 4. Approximation and artificial data
// Fixing the seed makes repeated runs use the same random innovations.
set_dynare_seed(2025);
// nograph suppresses figures; the computed IRFs remain in oo_.irfs.
stoch_simul(order=2, periods=1000, drop=100, irf=40, nograph);

// Export the observed series for the estimation example.
// verbatim passes ordinary MATLAB/Octave code through the Dynare parser.
// strcmp selects a row by name; the apostrophe makes a column of data.
// verbatim copies this block unchanged into the generated MATLAB/Octave
// program. It is not a Dynare equilibrium block; use % comments inside.
verbatim;
c1 = oo_.endo_simul(strcmp(M_.endo_names, 'c1'), :)';
k2 = oo_.endo_simul(strcmp(M_.endo_names, 'k2'), :)';
save('s03_simudata.mat', 'c1', 'k2');
end;
```

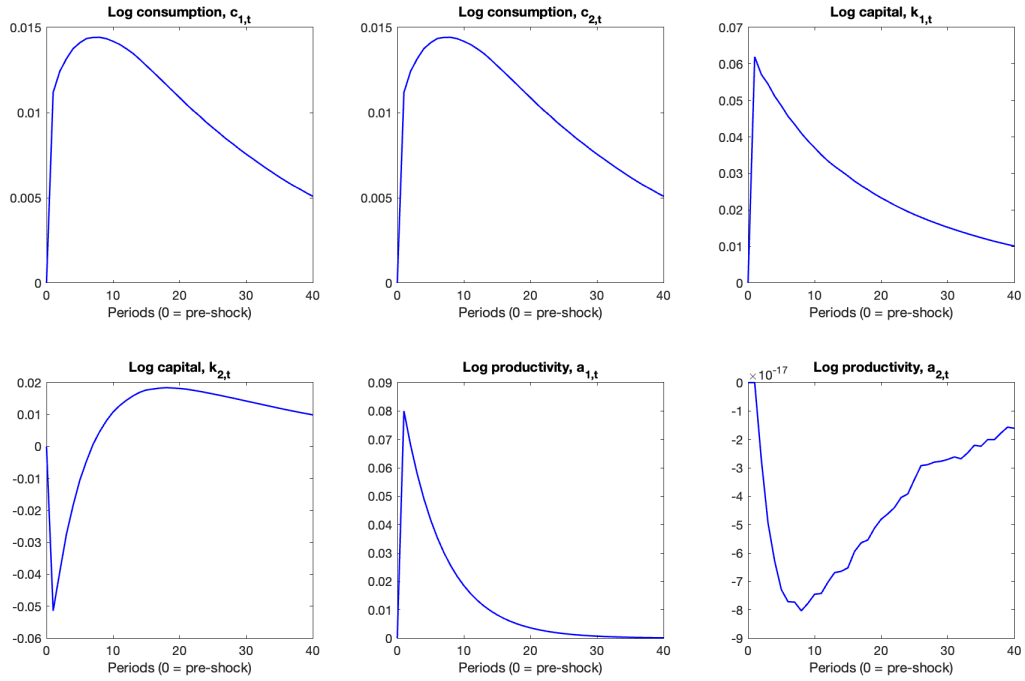


Figure 4: Two-country responses to a productivity innovation in country 1.
Notes: c_i, k_i, a_i are log consumption, log end-of-period capital and log productivity in country i. The innovation standard deviation is 0.08. Responses are changes in log units; period 0 is the pre-shock reference.

The coefficients can also be written as a consumption rule. Let $\widehat{k}_{i,t-1} = \ln K_{i,t-1} - \ln \bar{K}_i$ and $a_{i,t} = \ln A_{i,t}$ (steady-state productivity is one). At the calibration above, country 1's

second-order rule is

$$\begin{aligned}\ln C_{1,t} \simeq\ & 0.828143 + 0.243036(\widehat{k}_{1,t-1} + \widehat{k}_{2,t-1}) + 0.139989(a_{1,t} + a_{2,t}) \\ & + 0.061511(\widehat{k}^2_{1,t-1} + \widehat{k}^2_{2,t-1}) - 0.109951\widehat{k}_{1,t-1}\widehat{k}_{2,t-1} \\ & - 0.015489(a_{1,t}\widehat{k}_{1,t-1} + a_{2,t}\widehat{k}_{2,t-1}) \\ & - 0.032259(a_{1,t}\widehat{k}_{2,t-1} + a_{2,t}\widehat{k}_{1,t-1}) \\ & + 0.044325(a^2_{1,t} + a^2_{2,t}) - 0.040310a_{1,t}a_{2,t}.\end{aligned}$$

The left-hand side is log consumption, not its deviation from steady state. The constant includes the second-order uncertainty correction. Dynare reports lagged productivity and current innovations separately; substituting $a_{i,t} = 0.85a_{i,t-1} + \epsilon_{i,t}$ gives the compact form above. The quadratic coefficients already include the appropriate Taylor-expansion weights. At first order, both countries' capital and productivity raise country 1's consumption. Identical preferences and complete risk sharing imply the same consumption rule for country 2. Coefficients are rounded to six decimals.

Dynare also produces the associated impulse response functions of the system. Figure 4 illustrates the response of the system to a one-standard-deviation shock to ϵ_1.

The approximation file saves named observable series in a section-specific MAT file. Run it before the corresponding estimation example. The random seed is fixed for repeatability.

3.3 Estimation

Now assume that we do not know the true parameter values. We can estimate the unknown parameters from the saved data, using the Bayesian econometrics function of Dynare.

We will estimate 4 parameters $\rho, \alpha, \sigma_{\epsilon_1}$, and σ_{ϵ_2}. Hence, we specify a prior distribution for each parameter. The exercise uses beta priors for ρ and α, and finite-spread inverse-gamma priors for σ_{ϵ_1} and σ_{ϵ_2}. The Dynare code is as follows.

```
// ------------------------------------------------------------------------
// Section 3.3 | Bayesian estimation
// ------------------------------------------------------------------------
// 1. Variables and parameters
// All endogenous variables are logs. Countries share consumption under
// complete markets; each country has its own capital and productivity.
var
    c1    // Log consumption in country 1
    c2    // Log consumption in country 2
    k1    // Log capital chosen in country 1
    k2    // Log capital chosen in country 2
    a1    // Log productivity in country 1
    a2    // Log productivity in country 2
;
varexo
    e1    // Productivity innovation in country 1
    e2    // Productivity innovation in country 2
;

parameters gamma delta alpha beta rho;
```

```
gamma = 2;       // Consumption curvature
delta = 0.05;   // Depreciation rate
alpha = 0.4;    // Capital share
beta  = 0.98;   // Discount factor
rho   = 0.85;   // Productivity persistence

// 2. Equilibrium conditions
// exp(k1(-1)) and exp(k2(-1)) are the capital stocks used today.
model;
    // Complete risk sharing between identical households.
    c1 = c2;

    // Euler equations for investment in each country.
    exp(c1)^(-gamma) = beta * exp(c1(+1))^(-gamma)
        * (alpha*exp(a1(+1))*exp(k1)^(alpha-1) + 1-delta);

    exp(c2)^(-gamma) = beta * exp(c2(+1))^(-gamma)
        * (alpha*exp(a2(+1))*exp(k2)^(alpha-1) + 1-delta);

    // World output finances consumption and net capital accumulation.
    exp(c1) + exp(c2) + exp(k1) - (1-delta)*exp(k1(-1))
        + exp(k2) - (1-delta)*exp(k2(-1)) =
        exp(a1)*exp(k1(-1))^alpha + exp(a2)*exp(k2(-1))^alpha;

    // Country-specific productivity processes.
    a1 = rho*a1(-1) + e1;
    a2 = rho*a2(-1) + e2;
end;

// 3. Initial guesses in logs, and innovation standard deviations.
initval;
    k1 = 2.8;
    k2 = 2.8;
    c1 = 0.8;
    c2 = 0.8;
    a1 = 0;
    a2 = 0;
end;

shocks;
    var e1; stderr 0.08;
    var e2; stderr 0.08;
end;

steady;
check;

// 4. Bayesian estimation
// First run s03_02_twocountry_approx.mod in this folder.
// Each row gives a starting value, prior family, prior mean and prior SD.
varobs c1 k2;

estimated_params;
    rho, 0.85, beta_pdf, 0.84, 0.05;
```

```
    alpha, 0.4, beta_pdf, 0.38, 0.03;
    stderr e1, 0.08, inv_gamma_pdf, 0.078, 0.02;
    stderr e2, 0.08, inv_gamma_pdf, 0.082, 0.02;
end;

set_dynare_seed(2025);
// mode_compute=4 searches for the posterior mode before sampling.
// A short teaching chain; increase mh_replic for empirical inference.
estimation(datafile=s03_simudata, mode_compute=4, mh_replic=1000,
           mh_nblocks=1, nograph, nodisplay);
```

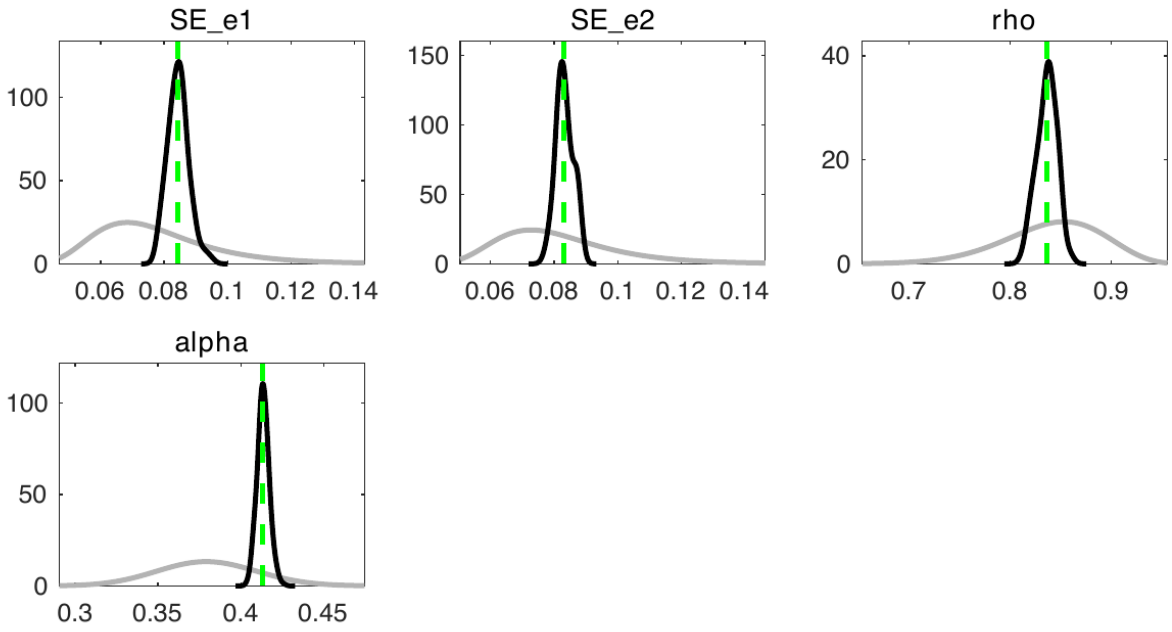


Figure 5: Prior and posterior distributions for the two-country model.
Notes: ρ is productivity persistence, α the capital share and $\sigma_{\epsilon_1}, \sigma_{\epsilon_2}$ the innovation standard deviations. Horizontal axes are parameter values and vertical axes are probability densities. The posterior uses simulated observations and a 1,000-draw teaching chain; convergence must be assessed before drawing empirical conclusions.

Given that there are two exogenous shock variables, we use two observables $\ln C_1$ and $\ln K_2$. Compare the posterior distributions from your estimation run with the priors and the parameter values used to generate the data.

4 Fiscal policies in the growth model

This section describes a suite of Dynare programs that implement the transition experiments performed in chapter 11 of Ljungqvist and Sargent (2004).

4.1 The model

Model 4.1: variables, fiscal instruments and baseline calibration.

Symbol	Code	Meaning / calibration
c_t	`c`	Consumption in goods units; endogenous.
k_{t-1}	`k(-1)`	Physical capital available at the beginning of period t; predetermined.
k_t	`k`	Capital chosen during period t for use in period $t+1$; endogenous.
r_t	`r`	Real rental yield on capital before tax and depreciation; endogenous.
w_t	`w`	Real wage per unit of labour; endogenous.
R_{t+1}	`R`	Forward consumption-adjusted gross saving return; endogenous.
s_t	`s`	Current after-tax capital payoff plus undepreciated capital; endogenous.
g_t	`g`	Exogenous government purchases; baseline 0.2.
$\tau_{c,t}$	`tauc`	Exogenous consumption tax rate; baseline 0.
$\tau_{k,t}$	`tauk`	Exogenous capital-income tax rate; baseline 0.
$\tau_{i,t}$	`taui`	Exogenous investment subsidy rate; purchase price $1-\tau_{i,t}$; baseline 0.
T_t	Residual closure	Lump-sum tax (negative for transfer) that balances fiscal spending net of distortionary receipts. Not an extra allocation equation.
β	`beta`	Discount factor: 0.95.
δ	`delta`	Depreciation rate: 0.2.
A	`A`	Productivity level: 1.
α	`alpha`	Capital share: 0.33.
γ	`gamma`	CRRA consumption curvature: 2.

Labour is fixed at one. All rates are per model period. The allocation variables and factor prices are endogenous; fiscal paths are announced exogenously. Steady-state guesses in `initval` are not calibration values.

A deterministic growth model has inelastic labor supply, an exogenous stream of government expenditures, and several kinds of distorting taxes. All variables are as described in chapter 11 of Ljungqvist and Sargent (2004). A representative agent maximizes

$$\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma}, \quad \beta \in (0,1)$$

Feasible allocations satisfy

$$g_t + c_t + k_t \leq A k_{t-1}^{\alpha} + (1-\delta) k_{t-1}.$$

Let T_t denote a residual lump-sum tax (negative for a transfer), which finances spending net of distorting tax receipts. This closes the fiscal budget without altering the marginal conditions. The household budget constraint is

$$\sum_{t=0}^{\infty} \{q_t (1+\tau_{ct}) c_t + (1-\tau_{it}) q_t [k_t - (1-\delta)k_{t-1}]\} \leq \sum_{t=0}^{\infty} q_t \{r_t (1-\tau_{kt}) k_{t-1} + w_t - T_t\}$$

and the government budget constraint is

$$\sum_{t=0}^{\infty} q_t g_t \leq \sum_{t=0}^{\infty} \{\tau_{ct} q_t c_t - \tau_{it} q_t [k_t - (1-\delta)k_{t-1}] + q_t r_t \tau_{kt} k_{t-1} + q_t T_t\}$$

The following conditions characterize an equilibrium:

$$c_t = A k_{t-1}^{\alpha} + (1-\delta)k_{t-1} - k_t - g_t \tag{7}$$
$$r_t = A\alpha k_{t-1}^{\alpha-1} \tag{8}$$
$$w_t = (1-\alpha) A k_{t-1}^{\alpha} \tag{9}$$
$$R_{t+1} = \frac{1+\tau_{c,t}}{1+\tau_{c,t+1}} \frac{(1-\tau_{i,t+1})(1-\delta) + (1-\tau_{k,t+1}) A\alpha k_t^{\alpha-1}}{1-\tau_{i,t}} \tag{10}$$
$$s_t = (1-\tau_{k,t}) r_t + 1 - \delta \tag{11}$$
$$c_t^{-\gamma} = \beta c_{t+1}^{-\gamma} R_{t+1}. \tag{12}$$

Interpreting factor prices and returns. The rental yield r_t is the amount of goods earned by one unit of capital during period t. The wage w_t is the goods earned per unit of labour. Labour is fixed at one in this model, so w_t is also total wage income.

A unit of capital provides rental income and leaves $1-\delta$ units of capital after depreciation. Its current after-tax payoff is therefore $s_t = (1-\tau_{k,t}) r_t + 1 - \delta$. The forward return R_{t+1} also accounts for the consumption-tax and investment-subsidy terms relevant to saving at t. Without taxes or subsidies, $R_{t+1} = 1 + r_{t+1} - \delta$.

Substituting the factor-price and return definitions into the Euler equation leaves two equations in c and k: the resource constraint and the Euler equation. The separate definitions make it easier to inspect wages and returns. In both the equations and the code, k_{t-1} is capital available at the beginning of t and k_t is chosen during t for use at $t+1$.

4.2 Parameter values

We set the parameters and the baseline values of the exogenous variables at values set in chapter 11 of Ljungqvist and Sargent (2004), as described in the following table are taken from RMT2.

Parameter/exogenous variable	Values
β	0.95
δ	0.2
A	1
α	0.33
γ	2
g	0.2
τ_c	0
τ_i	0
τ_k	0

Model 4.2: baseline fiscal calibration.
Notes: β is the discount factor, δ depreciation, A productivity, α the capital share and γ consumption curvature. g is government purchases; τ_c and τ_k are consumption and capital-income tax rates. τ_i is an investment subsidy, so its purchase-price wedge is $1 - \tau_i$. Rates are per model period; these are baseline policy levels, not innovations.

4.3 Transition experiments

4.3.1 A permanent increase in g.

The Dynare code below examines a permanent increase of 0.2 in g at t = 10.

The plots report real factor prices and capital payoffs.[1]

```
// ------------------------------------------------------------------------
// Section 4.3.1 | Permanent g increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    taui    // Investment subsidy rate (purchase price is 1-taui)
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
beta  = 0.95;    // Discount factor
```

[1]The 2010 notation r/q, w/q and s/q represents the real rental yield, wage and capital payoff. Here these quantities are calculated directly as r, w and s, so no intermediate q calculation is needed. The price q is introduced in the bond-price and yield exercise.

```
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 2;       // CRRA consumption curvature
// taui is an investment subsidy: the investment purchase price is 1-taui.

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (7) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (8) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (9) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (10) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * ((1-taui(+1))*(1-delta) + (1-tauk(+1))*A*alpha*k^(alpha-1))/(1-taui);

    // (11) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;

    // (12) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | g rises permanently at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    taui = 0;
    tauk = 0;
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.4;
    tauc = 0;
    taui = 0;
    tauk = 0;
```

```
end;

steady;

// Keep the old policy through t=9; endval applies from t=10 onward.
shocks(overwrite);
    var g;
    periods 1:10;
    values 0.2;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

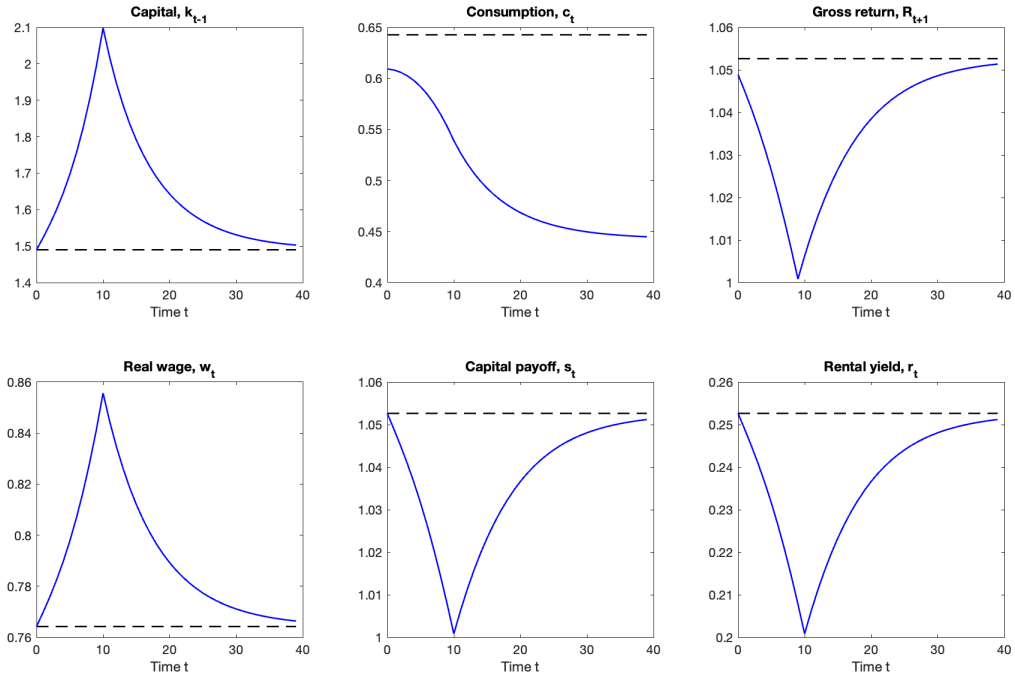


Figure 6: A permanent increase of g.
Notes: Government purchases rise permanently from 0.2 to 0.4 at $t = 10$, announced at $t = 0$. c_t is consumption, k_{t-1} capital available at the beginning of t, r_t the real rental yield, w_t the real wage, $s_t = (1 - \tau_{k,t})r_t + 1 - \delta$ the current capital payoff and R_{t+1} the forward gross return. Consumption curvature is $\gamma = 2$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

The following figure shows the path of the endogenous variables along the transition to the new steady state.
The remaining experiments each have a complete, independently runnable file.

4.3.2 A permanent increase in τ_c.

A permanent increase of τ_c at t = 10 of 20 percentage points.

```
// ------------------------------------------------------------------------
// Section 4.3.2 | Permanent tauc increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    taui    // Investment subsidy rate (purchase price is 1-taui)
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 2;       // CRRA consumption curvature
// taui is an investment subsidy: the investment purchase price is 1-taui.

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (7) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (8) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (9) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (10) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * ((1-taui(+1))*(1-delta) + (1-tauk(+1))*A*alpha*k^(alpha-1))/(1-taui);

    // (11) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;

    // (12) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;
end;
```

```
// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | tauc rises permanently at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    taui = 0;
    tauk = 0;
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.2;
    tauc = 0.2;
    taui = 0;
    tauk = 0;
end;

steady;

// Keep the old policy through t=9; endval applies from t=10 onward.
shocks(overwrite);
    var tauc;
    periods 1:10;
    values 0;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

4.3.3 A permanent increase in τ_i.

A permanent increase of τ_i at t = 10 of 20 percentage points.

```
// ------------------------------------------------------------------------
// Section 4.3.3 | Permanent taui increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
```

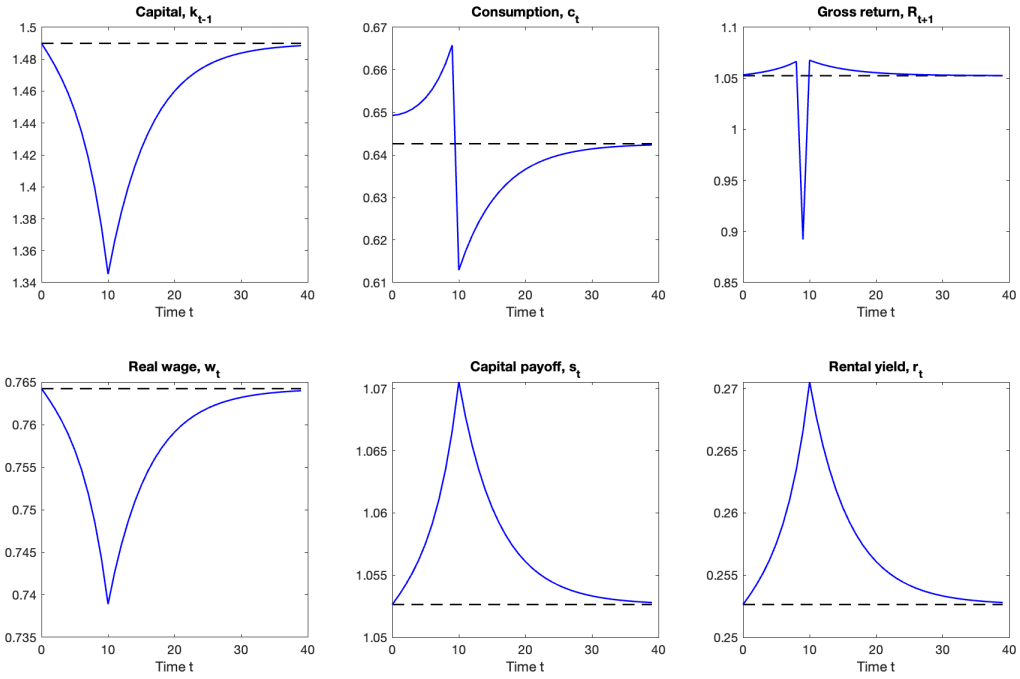


Figure 7: A permanent increase of τ_c.

Notes: The consumption-tax rate rises permanently from 0 to 0.2 at $t = 10$, announced at $t = 0$. c_t is consumption, k_{t-1} capital available at the beginning of t, r_t the real rental yield, w_t the real wage, $s_t = (1-\tau_{k,t})r_t + 1 - \delta$ the current capital payoff and R_{t+1} the forward gross return. Consumption curvature is $\gamma = 2$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

```
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    taui    // Investment subsidy rate (purchase price is 1-taui)
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 2;       // CRRA consumption curvature
// taui is an investment subsidy: the investment purchase price is 1-taui.

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (7) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (8) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (9) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (10) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * ((1-taui(+1))*(1-delta) + (1-tauk(+1))*A*alpha*k^(alpha-1))/(1-taui);

    // (11) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;

    // (12) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | taui rises permanently at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    taui = 0;
    tauk = 0;
```

```
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.2;
    tauc = 0;
    taui = 0.2;
    tauk = 0;
end;

steady;

// Keep the old policy through t=9; endval applies from t=10 onward.
shocks(overwrite);
    var taui;
    periods 1:10;
    values 0;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

4.3.4 A permanent increase in τ_k.

A permanent increase of τ_k at t = 10 of 20 percentage points.

```
// ------------------------------------------------------------------------
// Section 4.3.4 | Permanent tauk increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    taui    // Investment subsidy rate (purchase price is 1-taui)
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
```

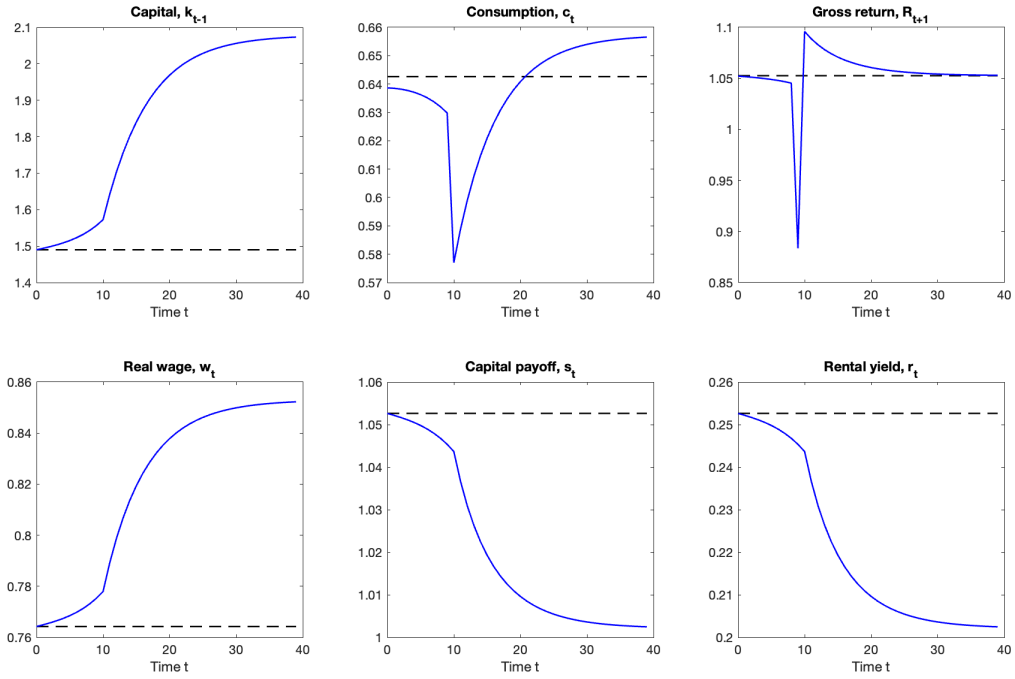


Figure 8: A permanent increase of τ_i.

Notes: The investment-subsidy rate rises permanently from 0 to 0.2 at $t = 10$, announced at $t = 0$. c_t is consumption, k_{t-1} capital available at the beginning of t, r_t the real rental yield, w_t the real wage, $s_t = (1-\tau_{k,t})r_t + 1 - \delta$ the current capital payoff and R_{t+1} the forward gross return. Consumption curvature is $\gamma = 2$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

```
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 2;       // CRRA consumption curvature
// taui is an investment subsidy: the investment purchase price is 1-taui.

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (7) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (8) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (9) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (10) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * ((1-taui(+1))*(1-delta) + (1-tauk(+1))*A*alpha*k^(alpha-1))/(1-taui);

    // (11) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;

    // (12) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | tauk rises permanently at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    taui = 0;
    tauk = 0;
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.2;
    tauc = 0;
    taui = 0;
```

```
    tauk = 0.2;
end;

steady;

// Keep the old policy through t=9; endval applies from t=10 onward.
shocks(overwrite);
    var tauk;
    periods 1:10;
    values 0;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

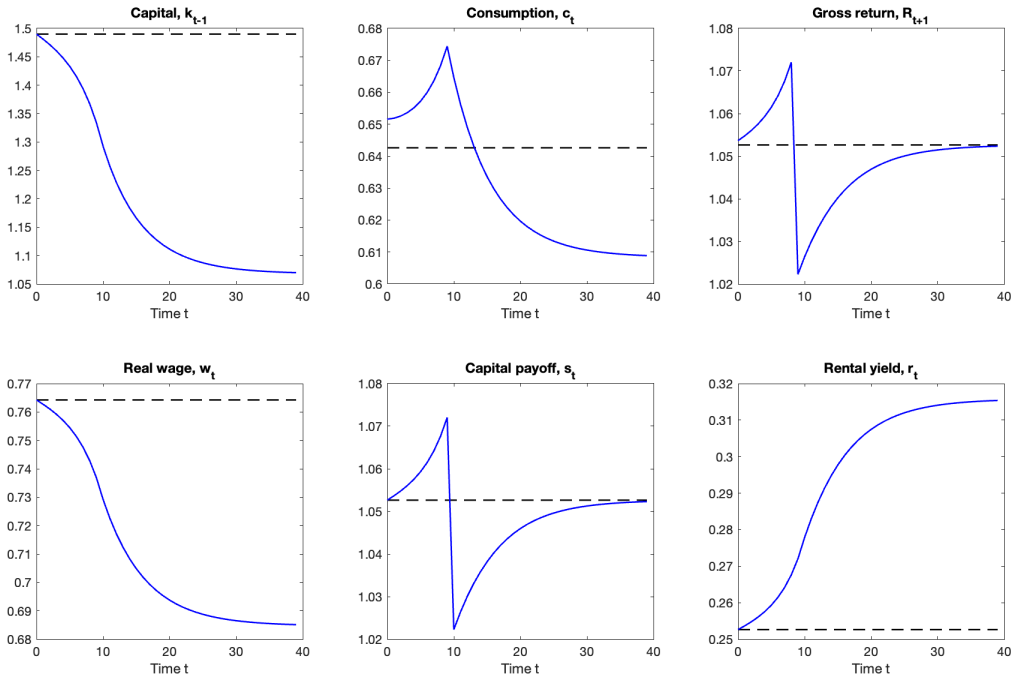


Figure 9: A permanent increase of τ_k.
Notes: The capital-income-tax rate rises permanently from 0 to 0.2 at $t = 10$, announced at $t = 0$. c_t is consumption, k_{t-1} capital available at the beginning of t, r_t the real rental yield, w_t the real wage, $s_t = (1-\tau_{k,t})r_t + 1 - \delta$ the current capital payoff and R_{t+1} the forward gross return. Consumption curvature is $\gamma = 2$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

4.3.5 One time increase in g.

A one time pulse of g at t = 10 of 0.2 .

```
// ------------------------------------------------------------------------
// Section 4.3.5 | One-period g increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    taui    // Investment subsidy rate (purchase price is 1-taui)
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 2;       // CRRA consumption curvature
// taui is an investment subsidy: the investment purchase price is 1-taui.

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (7) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (8) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (9) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (10) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * ((1-taui(+1))*(1-delta) + (1-tauk(+1))*A*alpha*k^(alpha-1))/(1-taui);

    // (11) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;

    // (12) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.
```

```
// 3. Scenario | g rises for one period at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    taui = 0;
    tauk = 0;
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// No endval block: the pulse returns to the initial steady state.
// One-period increase; all other periods keep the baseline value.
shocks(overwrite);
    var g;
    periods 11;
    values 0.4;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

4.3.6 One time increase in τ_i.

A one time pulse of τ_i at t = 10 of 20 percentage points.

```
// ------------------------------------------------------------------------
// Section 4.3.6 | One-period taui increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    taui    // Investment subsidy rate (purchase price is 1-taui)
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
```

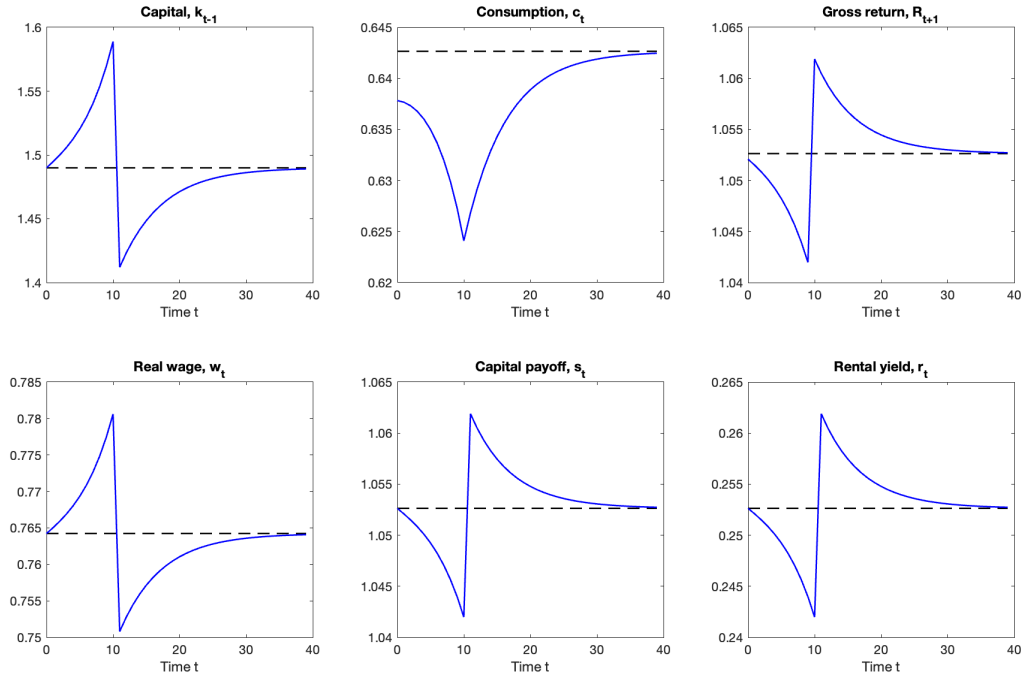


Figure 10: One time increase of g.

Notes: Government purchases rise from 0.2 to 0.4 only at $t = 10$, announced at $t = 0$. c_t is consumption, k_{t-1} capital available at the beginning of t, r_t the real rental yield, w_t the real wage, $s_t = (1 - \tau_{k,t})r_t + 1 - \delta$ the current capital payoff and R_{t+1} the forward gross return. Consumption curvature is $\gamma = 2$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

```
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 2;       // CRRA consumption curvature
// taui is an investment subsidy: the investment purchase price is 1-taui.

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (7) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (8) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (9) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (10) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * ((1-taui(+1))*(1-delta) + (1-tauk(+1))*A*alpha*k^(alpha-1))/(1-taui);

    // (11) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;

    // (12) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | taui rises for one period at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    taui = 0;
    tauk = 0;
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// No endval block: the pulse returns to the initial steady state.
// One-period increase; all other periods keep the baseline value.
shocks(overwrite);
    var taui;
    periods 11;
```

```
    values 0.2;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

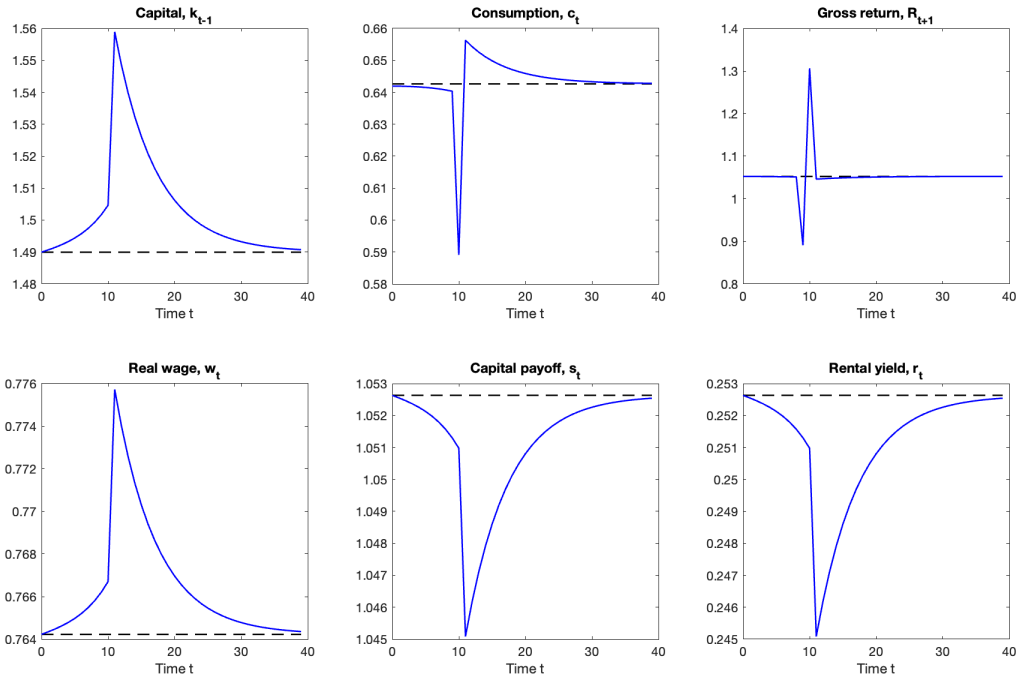


Figure 11: One time increase of τ_i.
Notes: The investment-subsidy rate rises from 0 to 0.2 only at $t = 10$, announced at $t = 0$. c_t is consumption, k_{t-1} capital available at the beginning of t, r_t the real rental yield, w_t the real wage, $s_t = (1 - \tau_{k,t})r_t + 1 - \delta$ the current capital payoff and R_{t+1} the forward gross return. Consumption curvature is $\gamma = 2$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

5 Fiscal policies in the growth model, 2.0

This section describes a suite of Dynare programs that implement the transition experiments performed in chapter 11 of Ljungqvist and Sargent (2012).

5.1 The model with inelastic labor supply

Model 5.1: variables, fiscal instruments and baseline calibration.

Symbol	Code	Meaning / calibration
c_t	`c`	Consumption in goods units; endogenous.
k_{t-1}	`k(-1)`	Physical capital available at the beginning of period t; predetermined.
k_t	`k`	Capital chosen during period t for use in period $t+1$; endogenous.
r_t	`r`	Real rental yield on capital before tax and depreciation; endogenous.
w_t	`w`	Real wage per unit of labour; endogenous.
R_{t+1}	`R`	Forward consumption-adjusted gross saving return; endogenous.
s_t	`s`	Current after-tax capital payoff plus undepreciated capital; endogenous.
g_t	`g`	Exogenous government purchases; baseline 0.2.
$\tau_{c,t}$	`tauc`	Exogenous consumption tax rate; baseline 0.
$\tau_{k,t}$	`tauk`	Exogenous capital-income tax rate; baseline 0.
T_t	Residual closure	Lump-sum tax (negative for transfer) that balances fiscal spending net of distortionary receipts. Not an extra allocation equation.
β	`beta`	Discount factor: 0.95.
δ	`delta`	Depreciation rate: 0.2.
A	`A`	Productivity level: 1.
α	`alpha`	Capital share: 0.33.
γ	`gamma`	CRRA consumption curvature: 2 (0.2 in the comparison economy).

Labour is fixed at one. Rates are per model period. Fiscal policies are exogenous announced paths; consumption, capital and real factor prices are endogenous. There is no investment subsidy in this model. Initial values for endogenous variables are numerical guesses.

A deterministic growth model has inelastic labor supply, an exogenous stream of government expenditures, and several kinds of distorting taxes. All variables are as described in chapter 11 of Ljungqvist and Sargent (2012). A representative agent maximizes

$$\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma}, \quad \beta \in (0,1)$$

Feasible allocations satisfy

$$g_t + c_t + k_t \leq A k_{t-1}^{\alpha} + (1-\delta) k_{t-1}$$

Let T_t denote a residual lump-sum tax (negative for a transfer), which finances spending net of distorting tax receipts. This closes the fiscal budget without altering the marginal conditions. The household budget constraint is

$$\sum_{t=0}^{\infty} \left\{ q_t c_t \left(1+\tau_{ct}\right) + q_t \left[k_t - (1-\delta) k_{t-1}\right] \right\} \leq \sum_{t=0}^{\infty} q_t \left\{ r_t \left(1-\tau_{kt}\right) k_{t-1} + w_t - T_t \right\}$$

and the government budget constraint is

$$\sum_{t=0}^{\infty} q_t g_t \leq \sum_{t=0}^{\infty} q_t \left\{\tau_{ct} c_t + \tau_{kt} r_t k_{t-1} + T_t\right\}$$

The following conditions characterize an equilibrium:

$$c_t = A k_{t-1}^{\alpha} + (1-\delta) k_{t-1} - k_t - g_t \tag{13}$$
$$r_t = A\alpha k_{t-1}^{\alpha-1} \tag{14}$$
$$w_t = (1-\alpha) A k_{t-1}^{\alpha} \tag{15}$$
$$R_{t+1} = \frac{1+\tau_{c,t}}{1+\tau_{c,t+1}} \left[1 - \delta + (1-\tau_{k,t+1}) A\alpha k_t^{\alpha-1}\right] \tag{16}$$
$$c_t^{-\gamma} = \beta c_{t+1}^{-\gamma} R_{t+1} \tag{17}$$
$$s_t = (1-\tau_{k,t}) r_t + 1 - \delta. \tag{18}$$

Each condition (13)–(18) appears explicitly in the Section 5.3 model blocks. Here r is the goods earned per unit of capital and w is the goods earned per unit of labour. The code `R` is R_{t+1}, the forward gross return on saving at date t. The current payoff s_t is reported separately for interpretation. Algebraic substitution of r, w, R, s reduces the model to two allocation equations in c and k. The notation matches the code: `k(-1)` is beginning-of-period k_{t-1} and `k` is the new choice k_t.

5.2 Parameter values

We set the parameters and the baseline values of the exogenous variables at values set in chapter 11 of Ljungqvist and Sargent (2012), as described in the following table are taken from RMT3.

Parameter/exogenous variable	Values
β	0.95
δ	0.2
A	1
α	0.33
γ	2
g	0.2
τ_c	0
τ_k	0

Model 5.2: baseline fiscal calibration.
Notes: β is the discount factor, δ depreciation, A productivity, α the capital share and γ consumption curvature. g is government purchases; τ_c and τ_k are consumption and capital-income tax rates. Rates are per model period; these are baseline policy levels, not innovations.

5.3 Transition experiments

5.3.1 Foreseen once-and-for-all increase in g.

The Dynare code below implements the experiment associated with Figure 11.6.1. which looks at the impact of a foreseen once-for-and-all increase of 0.2 in g at $t = 10$.

```
// ------------------------------------------------------------------------
// Section 5.3.1 | Permanent g increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 2;       // CRRA consumption curvature

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (13) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (14) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (15) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (16) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * (1-delta + (1-tauk(+1))*A*alpha*k^(alpha-1));

    // (17) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;

    // (18) Realized real capital payoff, distinct from forward R.
```

```
    s = (1-tauk)*r + 1-delta;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | g rises permanently at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    tauk = 0;
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.4;
    tauc = 0;
    tauk = 0;
end;

steady;

// Keep the old policy through t=9; endval applies from t=10 onward.
shocks(overwrite);
    var g;
    periods 1:10;
    values 0.2;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

The following figure shows the path of the endogenous variables along the transition to the new steady state.
The remaining experiments each have a complete, independently runnable file.

5.3.2 Foreseen once-and-for-all increase in τ_c.

A foreseen once-and-for-all increase in τ_c at $t = 10$ of 20 percentage points.(Figure 11.6.4)

```
// ------------------------------------------------------------------------
// Section 5.3.2 | Permanent tauc increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today.
var
```

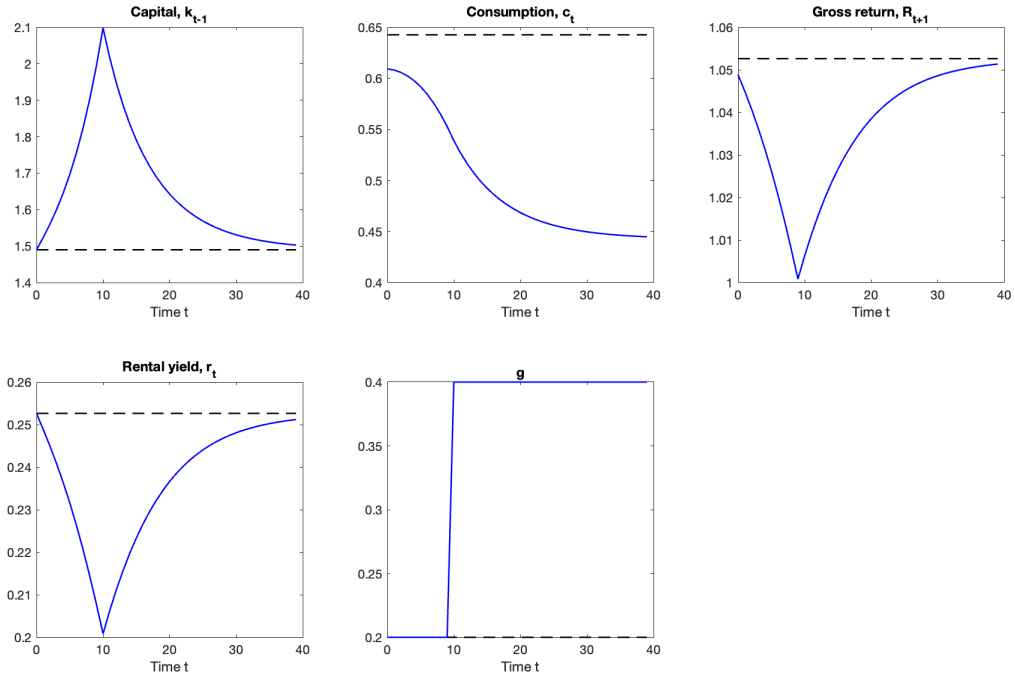


Figure 12: Foreseen once-and-for-all increase in g at $t = 10$.
Notes: Government purchases rise permanently from 0.2 to 0.4 at $t = 10$, announced at $t = 0$. c_t is consumption, k_{t-1} capital available at the beginning of t, r_t the real rental yield and R_{t+1} the forward gross return. Consumption curvature is $\gamma = 2$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

```
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 2;       // CRRA consumption curvature

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (13) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (14) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (15) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (16) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * (1-delta + (1-tauk(+1))*A*alpha*k^(alpha-1));

    // (17) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;

    // (18) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | tauc rises permanently at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    tauk = 0;
```

```
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.2;
    tauc = 0.2;
    tauk = 0;
end;

steady;

// Keep the old policy through t=9; endval applies from t=10 onward.
shocks(overwrite);
    var tauc;
    periods 1:10;
    values 0;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

5.3.3 Foreseen once-and-for-all increase in τ_k.

A foreseen once-and-for-all increase in τ_k at $t = 10$ of 20 percentage points.

```
// ------------------------------------------------------------------------
// Section 5.3.3 | Permanent tauk increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
```

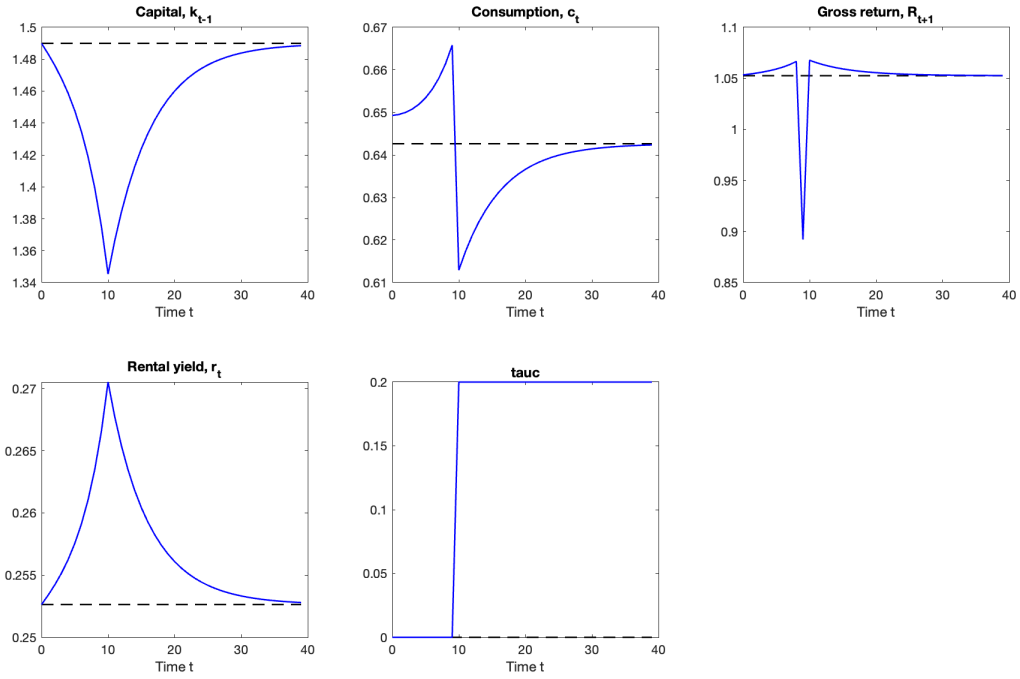


Figure 13: Foreseen once-and-for-all increase in τ_c at $t = 10$.
Notes: The consumption-tax rate rises permanently from 0 to 0.2 at $t = 10$, announced at $t = 0$. c_t is consumption, k_{t-1} capital available at the beginning of t, r_t the real rental yield and R_{t+1} the forward gross return. Consumption curvature is $\gamma = 2$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

```
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 2;       // CRRA consumption curvature

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (13) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (14) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (15) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (16) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * (1-delta + (1-tauk(+1))*A*alpha*k^(alpha-1));

    // (17) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;

    // (18) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | tauk rises permanently at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    tauk = 0;
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.2;
    tauc = 0;
    tauk = 0.2;
end;

steady;
```

```
// Keep the old policy through t=9; endval applies from t=10 onward.
shocks(overwrite);
    var tauk;
    periods 1:10;
    values 0;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

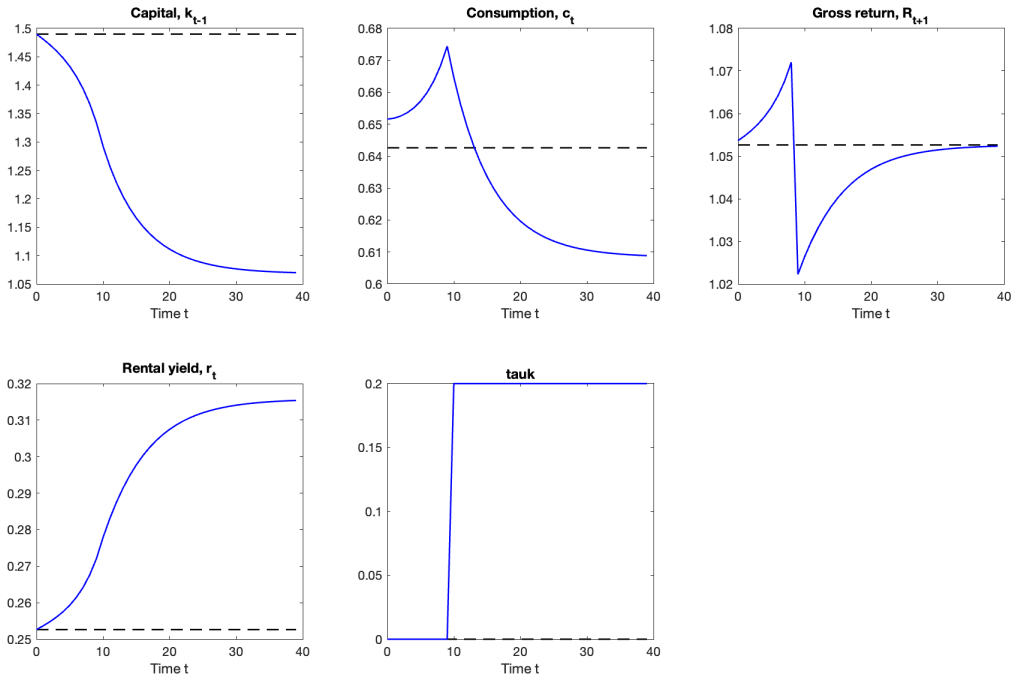


Figure 14: Foreseen once-and-for-all increase in τ_k at $t = 10$.
Notes: The capital-income-tax rate rises permanently from 0 to 0.2 at $t = 10$, announced at $t = 0$. c_t is consumption, k_{t-1} capital available at the beginning of t, r_t the real rental yield and R_{t+1} the forward gross return. Consumption curvature is $\gamma = 2$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

5.3.4 Foreseen one-time pulse increase in g.

A foreseen one-time pulse of g at $t = 10$ of 0.2 .

```
// ------------------------------------------------------------------------
// Section 5.3.4 | One-period g increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
```

```
// c: consumption; k: capital chosen today.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 2;       // CRRA consumption curvature

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (13) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (14) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (15) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (16) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * (1-delta + (1-tauk(+1))*A*alpha*k^(alpha-1));

    // (17) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;

    // (18) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | g rises for one period at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
```

```
    tauc = 0;
    tauk = 0;
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// No endval block: the pulse returns to the initial steady state.
// One-period increase; all other periods keep the baseline value.
shocks(overwrite);
    var g;
    periods 11;
    values 0.4;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

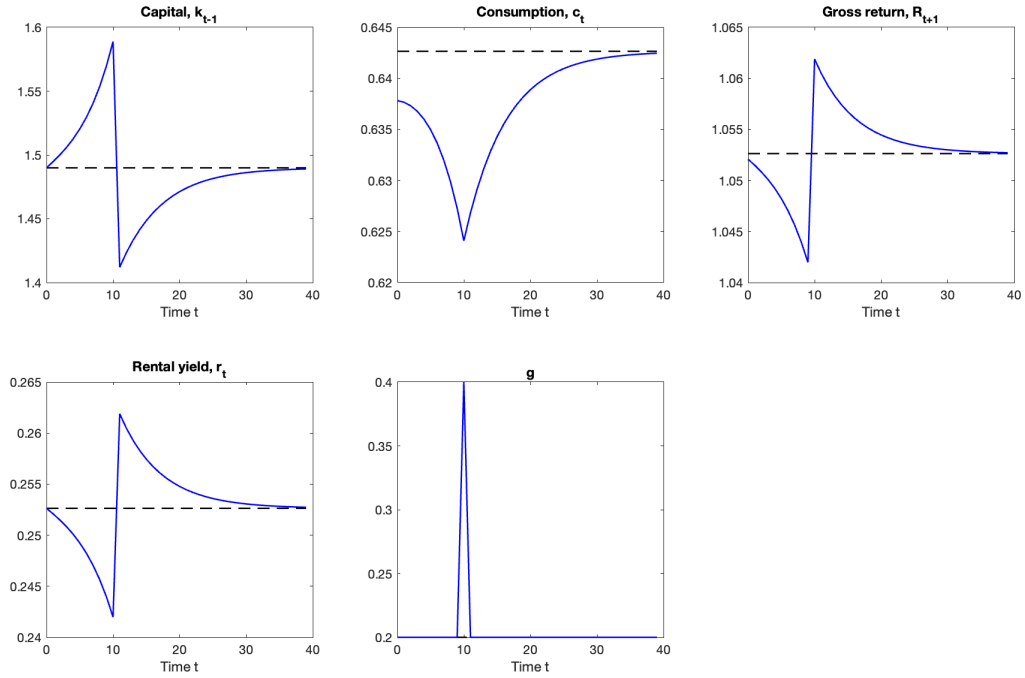


Figure 15: Foreseen one-time pulse increase in g at $t = 10$.
Notes: Government purchases rise from 0.2 to 0.4 only at $t = 10$, announced at $t = 0$. c_t is consumption, k_{t-1} capital available at the beginning of t, r_t the real rental yield and R_{t+1} the forward gross return. Consumption curvature is $\gamma = 2$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

5.3.5 Foreseen once-and-for-all increase g at t = 10 for two economies.

Responses to a foreseen increase in g at $t = 10$ for two economies, an economy with $\gamma = 2$ and an otherwise identical economy with $\gamma = .2$.

Compare the following low-curvature economy with `s05_03_01_g.mod` ($\gamma = 2$).

```
// ------------------------------------------------------------------------
// Section 5.3.5 | Permanent g increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 0.2;       // CRRA consumption curvature

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (13) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (14) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (15) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (16) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * (1-delta + (1-tauk(+1))*A*alpha*k^(alpha-1));

    // (17) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;

    // (18) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;
```

```
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | g rises permanently at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    tauk = 0;
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.4;
    tauc = 0;
    tauk = 0;
end;

steady;

// Keep the old policy through t=9; endval applies from t=10 onward.
shocks(overwrite);
    var g;
    periods 1:10;
    values 0.2;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;


// Compare this gamma=0.2 economy with s05_03_01_g.mod (gamma=2).
```

5.3.6 Bond prices and yields

How does an anticipated increase in government purchases change interest rates across maturities? This exercise uses the consumption response from Section 5.3.1 to answer that question.

A bond promises a consumption good at a future date. Let q_t be the price today of one good received at date t, and set $q_0 = 1$. If consumption is expected to be high at t, another good at that date is less valuable. The price also falls with the time until delivery because the household discounts future consumption.

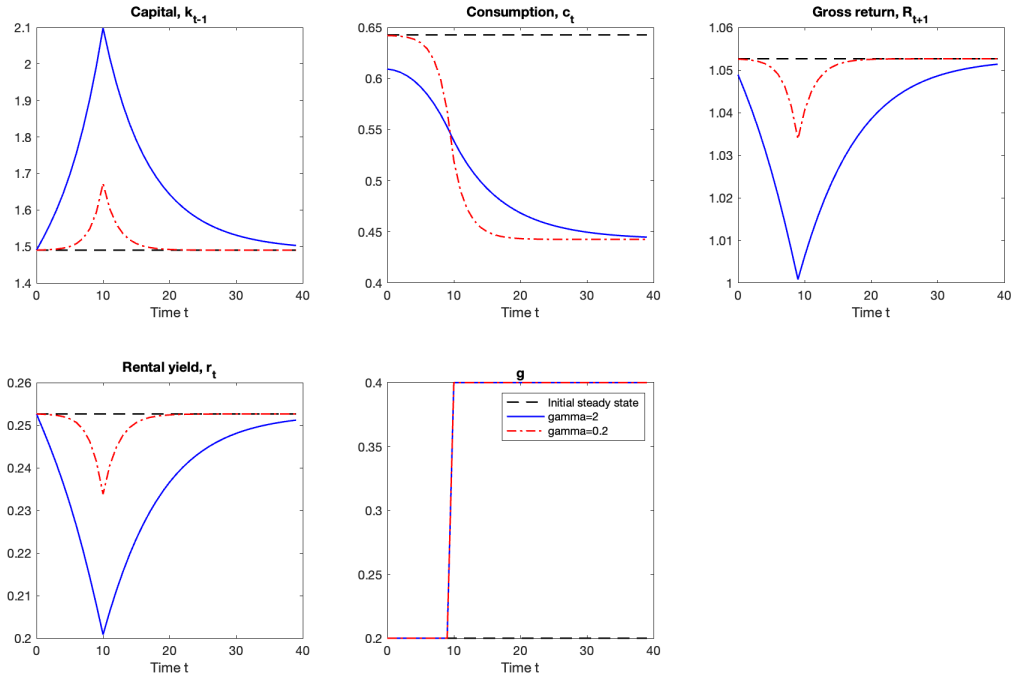


Figure 16: Response to foreseen once-and-for-all increase in g at $t = 10$. The solid line is for $\gamma = 2$, while the dashed-dotted line is for $\gamma = .2$.

Notes: Government purchases rise permanently from 0.2 to 0.4 at $t = 10$, announced at $t = 0$, in both economies. c_t is consumption, k_{t-1} capital available at the beginning of t, r_t the real rental yield and R_{t+1} the forward gross return. Solid blue: $\gamma = 2$; dash-dot red: $\gamma = 0.2$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

The solved consumption and tax paths give these prices directly:

$$q_t = \beta^t \left(\frac{c_t}{c_0} \right)^{-\gamma} \frac{1 + \tau_{c,0}}{1 + \tau_{c,t}}. \tag{19}$$

Here β is the discount factor, γ consumption curvature and $\tau_{c,t}$ the consumption-tax rate. The tax ratio is one in this government-spending experiment. No additional model equation is needed: this is a calculation using the consumption path already solved by Dynare.

At date t, a bond delivering one good i periods later costs q_{t+i}/q_t goods. Its continuously compounded yield per period is

$$y_{i,t} = -\frac{1}{i} \log \left(\frac{q_{t+i}}{q_t} \right), \qquad i \geq 1. \tag{20}$$

For example, a one-period bond costing 0.95 goods has gross return $1/0.95$ and continuously compounded yield $-\log(0.95)$. The exercise compares yield curves at dates $t = 0$, 10 and 60. Each curve shows the yields available at one date for repayment one to forty periods later. Calendar time and maturity are different: a forty-period bond bought at $t = 10$ repays at $t = 50$.

With constant consumption and taxes, $q_{t+i}/q_t = \beta^i$, so every maturity has yield $-\log(\beta)$. Expected consumption growth raises the one-period interest rate: saving must offer a higher return when consumption today is scarce relative to tomorrow. A longer-maturity yield is the average of the one-period log rates over the bond's life. Use the consumption panel to interpret the short-rate path, then explain how that path shapes the three yield curves.

```
// ------------------------------------------------------------------------
// Section 5.3.6 | Permanent g increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    tauk    // Capital-income tax rate
;

parameters beta delta A alpha gamma;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 2;       // CRRA consumption curvature
```

```
// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (13) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha + (1-delta)*k(-1) - k - g;

    // (14) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1);

    // (15) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha;

    // (16) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * (1-delta + (1-tauk(+1))*A*alpha*k^(alpha-1));

    // (17) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;

    // (18) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | g rises permanently at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    tauk = 0;
    k    = 1.5;
    c    = 0.4;
end;

steady;
check;

// Keep initial consumption for the reference line in the plot.
verbatim;
initial_cons = oo_.steady_state(strcmp(M_.endo_names, 'c'));
end;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.4;
    tauc = 0;
    tauk = 0;
end;

steady;
```

```
// Keep the old policy through t=9; endval applies from t=10 onward.
shocks(overwrite);
    var g;
    periods 1:10;
    values 0.2;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;


// Additional calculation for Section 5.3.6: dated prices and yields.
// This is MATLAB/Octave code, not another equilibrium equation.
// verbatim copies this block unchanged into the generated MATLAB/Octave
// program. It is not a Dynare equilibrium block; use % comments inside.
verbatim;
% M_.endo_names identifies the consumption row in Dynare's solution.
% Column 1 is the initial condition; columns 2:302 include the terminal date.
cons = oo_.endo_simul(strcmp(M_.endo_names, 'c'), 2:302);

% Parameters live in M_.params; select them by name rather than position.
discount = M_.params(strcmp(M_.param_names, 'beta'));
curvature = M_.params(strcmp(M_.param_names, 'gamma'));

% Normalize the date-zero Arrow-Debreu price to one, including tax wedges.
% The consumption tax is zero in this spending experiment; retaining it
% makes the pricing formula explicit if the scenario is later modified.
consumption_tax = oo_.exo_simul(2:302, strcmp(M_.exo_names, 'tauc'))';
q = discount.^(0:300) .* (cons/cons(1)).^(-curvature) ...
    .* (1+consumption_tax(1))./(1+consumption_tax);

% Select three observation dates; maturities are periods until repayment.
observation_dates = [0 10 60];
maturities = (1:40)';
yields = zeros(40, 3);
% Each column is one observation date; each row is one maturity.
% Add one because MATLAB index 1 represents model date t=0.
% These loops only calculate bond yields from the solved consumption path.
for curve = 1:3
    date_index = observation_dates(curve) + 1;
    for maturity = 1:40
        yields(maturity, curve) = ...
            -log(q(date_index+maturity)/q(date_index))/maturity;
    end
end

short_rate = -log(q(2:end)./q(1:end-1));

% Calendar time belongs on the first two panels; maturity on the third.
time = (0:39)';
figure('Position', [100 100 1200 400]);
subplot(1, 3, 1);
```

```
plot(time, initial_cons*ones(40, 1), '--k', time, cons(1:40), 'b');
title('Consumption'); xlabel('Time t'); ylabel('Goods');
legend('Initial steady state', 'Spending increase');

subplot(1, 3, 2);
plot(time, -log(discount)*ones(40, 1), '--k', ...
     time, short_rate(1:40), 'b');
title('One-period interest rate'); xlabel('Time t'); ylabel('Log rate per period');

subplot(1, 3, 3);
plot(maturities, yields(:, 1), 'b', maturities, yields(:, 2), '-.r', ...
     maturities, yields(:, 3), '--k');
title('Yield curves'); xlabel('Maturity (periods)'); ylabel('Log yield per period');
legend('t=0', 't=10', 't=60');
end;
```

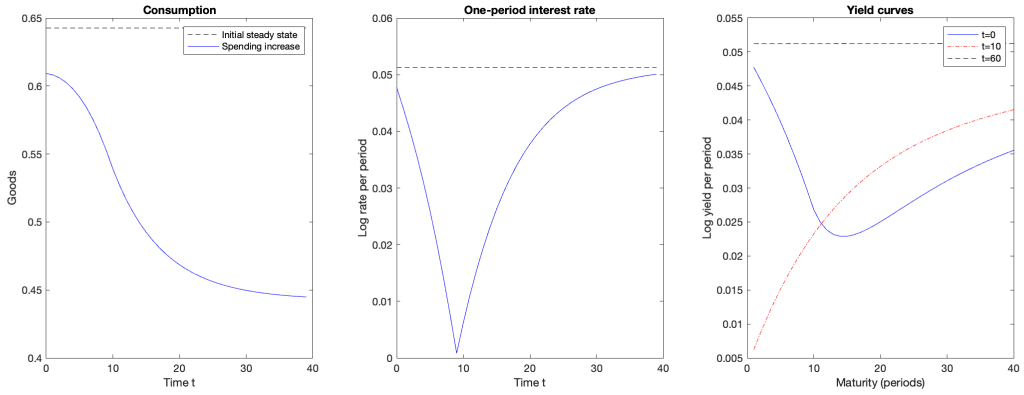


Figure 17: Consumption, interest rates and the yield curve after an announced spending increase.

Notes: Government purchases rise permanently from 0.2 to 0.4 at $t = 10$, announced at $t = 0$. The first two panels show calendar time; the third shows maturities of 1–40 periods at observation dates $t = 0$ (solid blue), $t = 10$ (dash-dot red) and $t = 60$ (dashed black). The one-period rate and bond yields are continuously compounded, per model period. Dashed black lines in the first two panels show the initial steady state; its flat yield curve is $-\log(0.95)$. Calibration: $\beta = 0.95$, $\delta = 0.2$, $A = 1$, $\alpha = 0.33$, $\gamma = 2$; tax rates are zero.

5.4 The model with elastic labor supply

Model 5.4: variables, fiscal instruments and baseline calibration.

Symbol	Code	Meaning / calibration
c_t	`c`	Consumption in goods units; endogenous.
k_{t-1}	`k(-1)`	Physical capital available at the beginning of period t; predetermined.
k_t	`k`	Capital chosen during period t for use in period $t+1$; endogenous.

Symbol	Code	Meaning / calibration (continued)
n_t	`n`	Hours as a fraction of available time, between zero and one; endogenous.
r_t	`r`	Real rental yield on capital before tax and depreciation; endogenous.
w_t	`w`	Real wage per unit of labour; endogenous.
R_{t+1}	`R`	Forward consumption-adjusted gross saving return; endogenous.
s_t	`s`	Current after-tax capital payoff plus undepreciated capital; endogenous.
g_t	`g`	Exogenous government purchases; baseline 0.2.
$\tau_{c,t}$	`tauc`	Exogenous consumption tax rate; baseline 0.
$\tau_{k,t}$	`tauk`	Exogenous capital-income tax rate; baseline 0.
T_t	Residual closure	Lump-sum tax (negative for transfer) that balances fiscal spending net of distortionary receipts. Not an extra allocation equation.
$\tau_{n,t}$	`taun`	Exogenous labour-income tax rate; baseline 0.
β	`beta`	Discount factor: 0.95.
δ	`delta`	Depreciation rate: 0.2.
A	`A`	Productivity level: 1.
α	`alpha`	Capital share: 0.33.
γ	`gamma`	Consumption curvature: 1, implementing log utility.
B	`B`	Weight on leisure: 3.

This model uses log consumption and linear leisure utility. Rates are per model period. Labour is endogenous and wage income is $w_t n_t$. Fiscal paths are exogenous. Initial values are solver guesses; the equilibrium must have $c, k > 0$ and $0 < n < 1$.

In this section we modify the previous model by allowing for elastic labor supply. Now the representative agent maximizes

$$\sum_{t=0}^{\infty} \beta^t [\log c_t + B(1 - n_t)], \quad \beta \in (0, 1)$$

where B is substantially greater than 1 to assure an interior solution $n \in (0, 1)$ for labor supply. Feasible allocations satisfy

$$g_t + c_t + k_t \leq A k_{t-1}^{\alpha} n_t^{1-\alpha} + (1 - \delta) k_{t-1}.$$

Let T_t denote a residual lump-sum tax (negative for a transfer), which finances spending net of distorting tax receipts. This closes the fiscal budget without altering the marginal conditions. The household budget constraint is

$$\sum_{t=0}^{\infty} \{q_t (1 + \tau_{ct}) c_t + q_t [k_t - (1 - \delta) k_{t-1}]\} \leq \sum_{t=0}^{\infty} q_t \{r_t (1 - \tau_{kt}) k_{t-1} + w_t n_t (1 - \tau_{nt}) - T_t\}$$

and the government budget constraint is

$$\sum_{t=0}^{\infty} q_t g_t \leq \sum_{t=0}^{\infty} q_t \left\{\tau_{ct} c_t + \tau_{kt} r_t k_{t-1} + \tau_{nt} n_t w_t + T_t\right\}$$

The following conditions characterize the endogenous variables in the model equilibrium:

$$c_t = A k_{t-1}^{\alpha} n_t^{1-\alpha} + (1-\delta) k_{t-1} - k_t - g_t \tag{21}$$
$$r_t = A\alpha k_{t-1}^{\alpha-1} n_t^{1-\alpha} \tag{22}$$
$$w_t = A(1-\alpha) k_{t-1}^{\alpha} n_t^{-\alpha} \tag{23}$$
$$R_{t+1} = \frac{1+\tau_{c,t}}{1+\tau_{c,t+1}} \left[1 - \delta + (1-\tau_{k,t+1}) A\alpha k_t^{\alpha-1} n_{t+1}^{1-\alpha}\right] \tag{24}$$
$$s_t = (1-\tau_{k,t}) r_t + 1 - \delta \tag{25}$$
$$c_t^{-1} = \beta c_{t+1}^{-1} R_{t+1} \tag{26}$$
$$B c_t (1+\tau_{c,t}) = (1-\tau_{n,t}) w_t. \tag{27}$$

Every condition (21)–(27) is active in the Section 5.4 listings. Here `gamma=1` implements log consumption utility. The wage w_t is per hour, so wage income is $w_t n_t$; r_t is the rental yield before depreciation, s_t a current capital payoff, and R_{t+1} the forward gross return represented by code `R`. In both text and code, beginning-of-period capital is k_{t-1} (`k(-1)`), while k_t (`k`) is chosen during t for use at $t+1$. Eliminating the price and return definitions reduces the system to resource, Euler and labour-supply equations in c, k, n.

The wage and rental yield are measured in goods.

5.4.1 Elastic Labor supply : Unforseen once-and-for-all increase in g at t = 0.

An unforeseen once-and-for-all increase in g from 0.2 to 0.4 at $t = 0$.

```
// -------------------------------------------------------------------------
// Section 5.4.1 | Permanent g increase
// -------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today. n: hours worked.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    n       // Labour supplied, n_t
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    tauk    // Capital-income tax rate
    taun    // Labour-income tax rate
```

```
;

parameters beta delta A alpha gamma B;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 1;       // Log-consumption utility
B     = 3;       // Weight on leisure

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (21) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha*n^(1-alpha) + (1-delta)*k(-1) - k - g;

    // (22) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1)*n^(1-alpha);

    // (23) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha*n^(-alpha);

    // (24) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * (1-delta + (1-tauk(+1))*A*alpha*k^(alpha-1)*n(+1)^(1-alpha));

    // (25) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;

    // (26) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;

    // (27) Consumption-leisure condition with the consumption-tax wedge.
    B*c*(1+tauc) = (1-taun)*w;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | g rises permanently at text date t=0
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    tauk = 0;
    taun = 0;
    k    = 1.5;
    c    = 0.4;
    n    = 0.4;
end;

steady;
```

```
check;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.4;
    tauc = 0;
    tauk = 0;
    taun = 0;
end;

steady;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

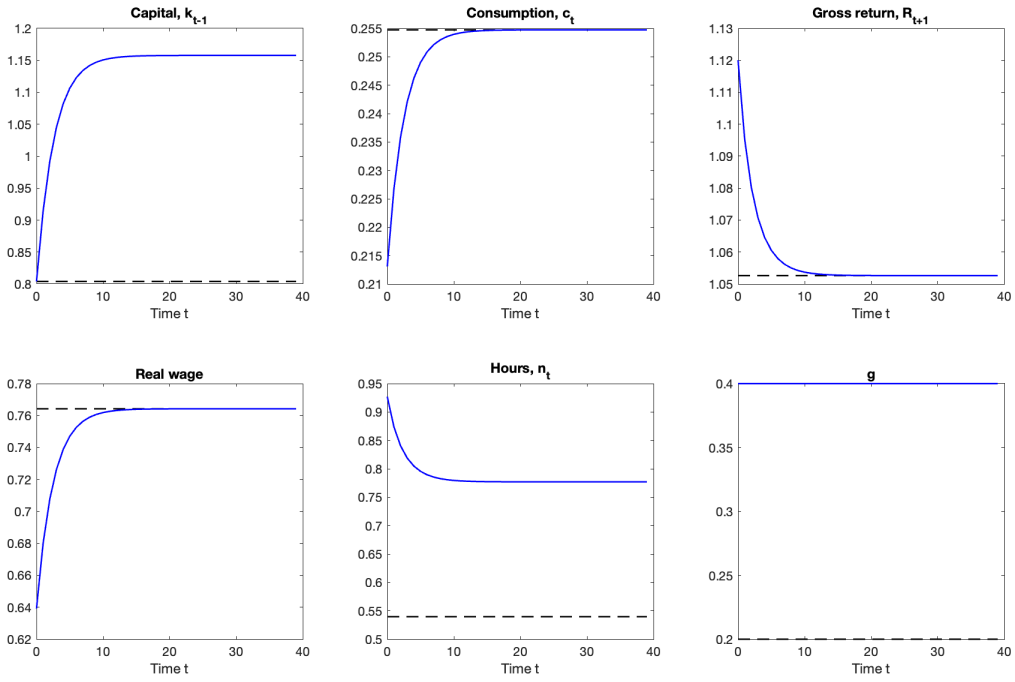


Figure 18: Elastic labor supply: response to unforeseen increase in g at $t = 0$.
Notes: Government purchases rise permanently from 0.2 to 0.4 at $t = 0$, with no earlier anticipation. c_t is consumption, k_{t-1} capital available at the beginning of t, w_t the real wage and R_{t+1} the forward gross return. n_t is hours; wage income is $w_t n_t$. Preferences use $\gamma = 1$ and $B = 3$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

5.4.2 Elastic Labor supply : Unforseen once-and-for-all increase in τ_n at t = 0.

An unforeseen once-and-for-all increase in τ_n to 20 percentage points at $t = 0$.

```
// ------------------------------------------------------------------------
// Section 5.4.2 | Permanent taun increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today. n: hours worked.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
    n       // Labour supplied, n_t
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    tauk    // Capital-income tax rate
    taun    // Labour-income tax rate
;

parameters beta delta A alpha gamma B;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 1;       // Log-consumption utility
B     = 3;       // Weight on leisure

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (21) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha*n^(1-alpha) + (1-delta)*k(-1) - k - g;

    // (22) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1)*n^(1-alpha);

    // (23) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha*n^(-alpha);

    // (24) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * (1-delta + (1-tauk(+1))*A*alpha*k^(alpha-1)*n(+1)^(1-alpha));

    // (25) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;
```

```
    // (26) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;

    // (27) Consumption-leisure condition with the consumption-tax wedge.
    B*c*(1+tauc) = (1-taun)*w;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | taun rises permanently at text date t=0
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
initval;
    g    = 0.2;
    tauc = 0;
    tauk = 0;
    taun = 0;
    k    = 1.5;
    c    = 0.4;
    n    = 0.4;
end;

steady;
check;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.2;
    tauc = 0;
    tauk = 0;
    taun = 0.2;
end;

steady;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

5.4.3 Elastic Labor supply : Foreseen once-and-for-all increase in τ_n at t = 10.

A foreseen once-and-for-all increase in τ_n to 20 percentage points at $t = 10$.

```
// ------------------------------------------------------------------------
// Section 5.4.3 | Permanent taun increase
// ------------------------------------------------------------------------
// 1. Variables and calibration
// c: consumption; k: capital chosen today. n: hours worked.
var
    c       // Consumption, c_t
    k       // Capital chosen during t, k_t (available in t+1)
```

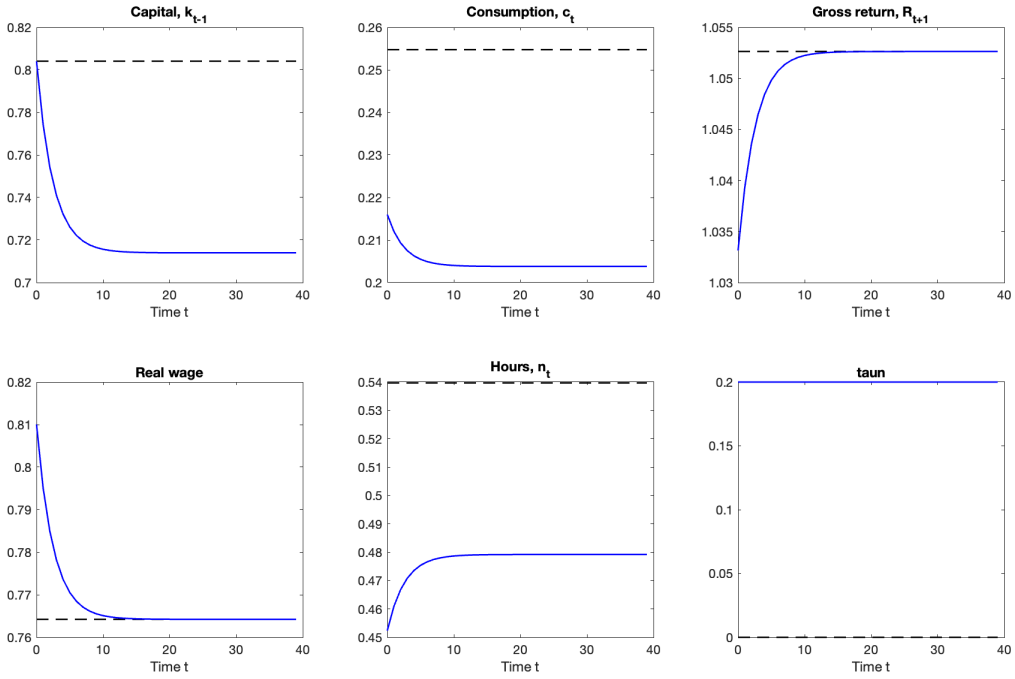


Figure 19: Elastic labor supply: response to unforeseen increase in τ_n at $t = 0$.
Notes: The labour-income-tax rate rises permanently from 0 to 0.2 at $t = 0$, with no earlier anticipation. c_t is consumption, k_{t-1} capital available at the beginning of t, w_t the real wage and R_{t+1} the forward gross return. n_t is hours; wage income is $w_t n_t$. Preferences use $\gamma = 1$ and $B = 3$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

```
    n       // Labour supplied, n_t
    r       // Current real rental yield, r_t (before depreciation)
    w       // Current real wage, w_t
    R       // Forward consumption-adjusted gross return, R_(t+1)
    s       // Current after-tax capital payoff, s_t
;
varexo
    g       // Government purchases
    tauc    // Consumption tax rate
    tauk    // Capital-income tax rate
    taun    // Labour-income tax rate
;

parameters beta delta A alpha gamma B;
beta  = 0.95;    // Discount factor
delta = 0.2;     // Depreciation rate
A     = 1;       // Productivity level
alpha = 0.33;    // Capital share
gamma = 1;       // Log-consumption utility
B     = 3;       // Weight on leisure

// 2. Equilibrium conditions: one equation per displayed condition.
// Text and code share Dynare timing: k(-1) is k_(t-1); k is k_t.
model;
    // (21) Resource constraint; k(-1) is capital available today.
    c = A*k(-1)^alpha*n^(1-alpha) + (1-delta)*k(-1) - k - g;

    // (22) Current real rental yield (before taxes and depreciation).
    r = A*alpha*k(-1)^(alpha-1)*n^(1-alpha);

    // (23) Current real wage; with n=1 it is also total labour income.
    w = A*(1-alpha)*k(-1)^alpha*n^(-alpha);

    // (24) Forward gross return R_(t+1) on saving at date t.
    // Capital chosen today earns the rental yield in period t+1.
    R = (1+tauc)/(1+tauc(+1))
        * (1-delta + (1-tauk(+1))*A*alpha*k^(alpha-1)*n(+1)^(1-alpha));

    // (25) Realized real capital payoff, distinct from forward R.
    s = (1-tauk)*r + 1-delta;

    // (26) Consumption Euler equation.
    c^(-gamma) = beta*c(+1)^(-gamma)*R;

    // (27) Consumption-leisure condition with the consumption-tax wedge.
    B*c*(1+tauc) = (1-taun)*w;
end;

// Eliminating r, w, R and s reduces this to the allocation model.
// The separate definitions also give the wages and returns to inspect.

// 3. Scenario | taun rises permanently at text date t=10
// Policy is known at t=0. Simulation period 1 corresponds to text t=0.
// Initial steady state: guesses for c and k are refined by steady.
```

```
initval;
    g    = 0.2;
    tauc = 0;
    tauk = 0;
    taun = 0;
    k    = 1.5;
    c    = 0.4;
    n    = 0.4;
end;

steady;
check;

// Terminal steady state under the permanent new policy.
endval;
    g    = 0.2;
    tauc = 0;
    tauk = 0;
    taun = 0.2;
end;

steady;

// Keep the old policy through t=9; endval applies from t=10 onward.
shocks(overwrite);
    var taun;
    periods 1:10;
    values 0;
end;

// 4. Solve the transition; a long horizon lets the path settle down.
perfect_foresight_setup(periods=300);
perfect_foresight_solver;
```

6 A full-scale fiscal RBC model

This section is reserved for the flexible-price version of the separate NT-DSGE model. Its implementation and exercises will be added separately.

A. List of Dynare Files

The maintained participant files are in `Code/sections`; self-contained section-output files are in `Code/full_replications`. Run each approximation before its estimation file in the same directory to create the named artificial data. Each full-output file generates its own required data. The files below are the examples used in this workbook.

Participant examples.

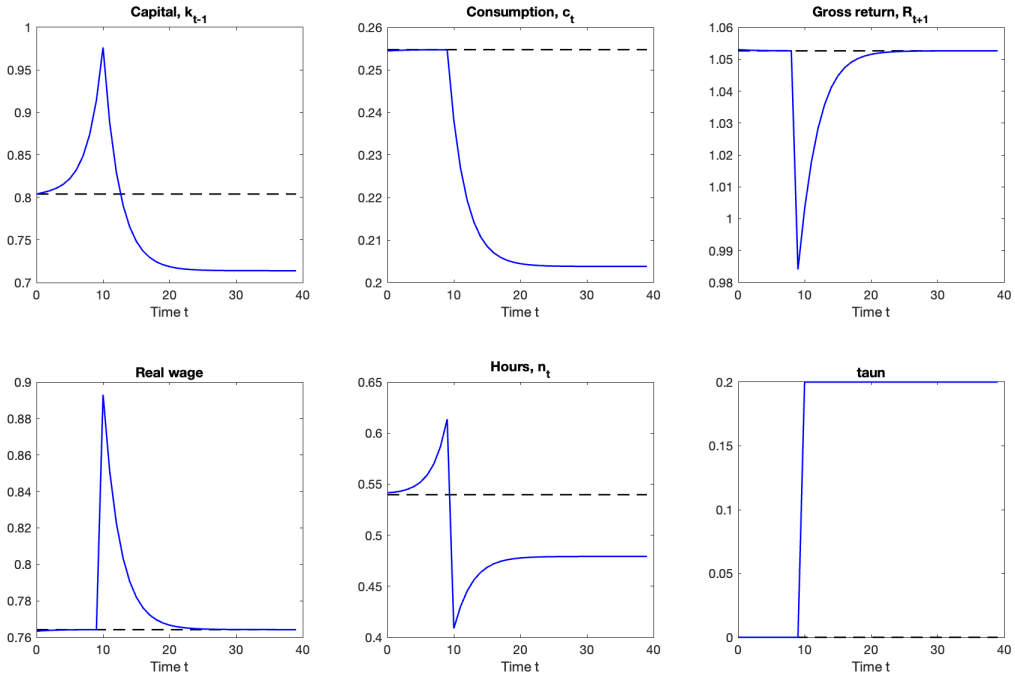


Figure 20: Elastic labor supply: response to foreseen increase in τ_n at $t = 10$.
Notes: The labour-income-tax rate rises permanently from 0 to 0.2 at $t = 10$, announced at $t = 0$. c_t is consumption, k_{t-1} capital available at the beginning of t, w_t the real wage and R_{t+1} the forward gross return. n_t is hours; wage income is $w_t n_t$. Preferences use $\gamma = 1$ and $B = 3$. The initial policy is $g = 0.2$ with tax/subsidy rates zero; $\beta = 0.95$, $\delta = 0.2$, $A = 1$ and $\alpha = 0.33$. Dashed black lines show initial steady-state values. All quantities and rates are per model period.

- `Code/sections/s02_02_growth_approx.mod` — Section 2.2: approximate and simulate the one-sector growth model.
- `Code/sections/s02_03_growth_estimate.mod` — Section 2.3: estimate the one-sector model from artificial consumption data.
- `Code/sections/s03_02_twocountry_approx.mod` — Section 3.2: approximate and simulate the two-country growth model.
- `Code/sections/s03_03_twocountry_estimate.mod` — Section 3.3: estimate the two-country model from artificial observations.
- `Code/sections/s04_03_01_g.mod` — 4.3.1: permanent government-purchases increase.
- `Code/sections/s04_03_02_tauc.mod` — 4.3.2: permanent consumption-tax increase.
- `Code/sections/s04_03_03_taui.mod` — 4.3.3: permanent investment-subsidy increase.
- `Code/sections/s04_03_04_tauk.mod` — 4.3.4: permanent capital-income-tax increase.
- `Code/sections/s04_03_05_g.mod` — 4.3.5: one-period government-purchases pulse.
- `Code/sections/s04_03_06_taui.mod` — 4.3.6: one-period investment-subsidy pulse.
- `Code/sections/s05_03_01_g.mod` — 5.3.1: anticipated permanent government-purchases increase.
- `Code/sections/s05_03_02_tauc.mod` — 5.3.2: anticipated permanent consumption-tax increase.
- `Code/sections/s05_03_03_tauk.mod` — 5.3.3: anticipated permanent capital-income-tax increase.
- `Code/sections/s05_03_04_g.mod` — 5.3.4: anticipated one-period government-purchases pulse.
- `Code/sections/s05_03_05_g_gamma02.mod` — 5.3.5: low-curvature comparison economy.
- `Code/sections/s05_03_06_yields.mod` — 5.3.6: dated prices and yield calculations.
- `Code/sections/s05_04_01_g.mod` — 5.4.1: immediate permanent government-purchases increase with elastic labour.
- `Code/sections/s05_04_02_taun.mod` — 5.4.2: immediate permanent labour-income-tax increase.
- `Code/sections/s05_04_03_taun.mod` — 5.4.3: anticipated permanent labour-income-tax increase.

Self-contained full-output examples.

- `Code/full_replications/s02_full_rep.mod` — Section 2: simulation, estimation and IRF/density outputs.
- `Code/full_replications/s03_full_rep.mod` — Section 3: simulation, estimation and IRF/density outputs.
- `Code/full_replications/s04_full_rep.mod` — All six Section 4 transition experiments and panels.
- `Code/full_replications/s05_03_full_rep.mod` — Section 5.3 policy experiments, curvature comparison and yields.
- `Code/full_replications/s05_04_full_rep.mod` — All three Section 5.4 labour-supply experiments and panels.

References

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Kim, J. and S. H. Kim (2003, August). Spurious welfare reversals in international business cycle models. Journal of International Economics 60(2), 471-500.

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Siow, A. (1984, May). Occupational choice under uncertainty. Econometrica 52(3), 631-45.